

Fire developmental atlas: geometric life histories and environmental transitions

WHAT THE CURRENT DATA CLEARLY SHOW

25 fires and 204 usable observation-to-observation transitions are available. Evidence: strong descriptive evidence.

Task-specific QC is required: morphology is usable for 25 fires, pulse detection for 25, and terminal-observation analysis for 8. Evidence: strong descriptive evidence.

Final meaningful expansion, terminal observed record, and physical extinction are distinct quantities. Physical extinction is not observed. Evidence: strong descriptive evidence.

WHAT THE CURRENT DATA SUGGEST

The most coherent geometry-centered clock in this sample is peak expansion. Evidence: moderate descriptive evidence from alignment ranking.

Fires show pulses, pauses, repeated expansions, and prolonged low-growth intervals rather than a single uniform terminal pattern. Evidence: moderate descriptive evidence from profiles and event table.

Discrete geometry-only groups have suggestive support; best silhouette is 0.48. Continuous axes should be considered alongside clusters.

WHAT THE CURRENT DATA DO NOT SUPPORT

A common terminal climate endpoint is unsupported with current data; terminal observed records remain heterogeneous.

Causal claims about wind, VPD, or temperature triggering expansion are not supported with 25 daily fire sequences.

No prescribed-fire comparison is made or implied.

WHAT SHOULD BE ANALYZED NEXT

Expand the event sample and reduce temporal gaps before physical cessation claims.

Compute climate support on newly burned area and advancing boundaries, not only nearest grid cell to event centroid.

Use daily wind speed here only as a scalar first pass; direction, gusts, and subdaily wind remain future work.

WHAT COULD BECOME THE MANUSCRIPT

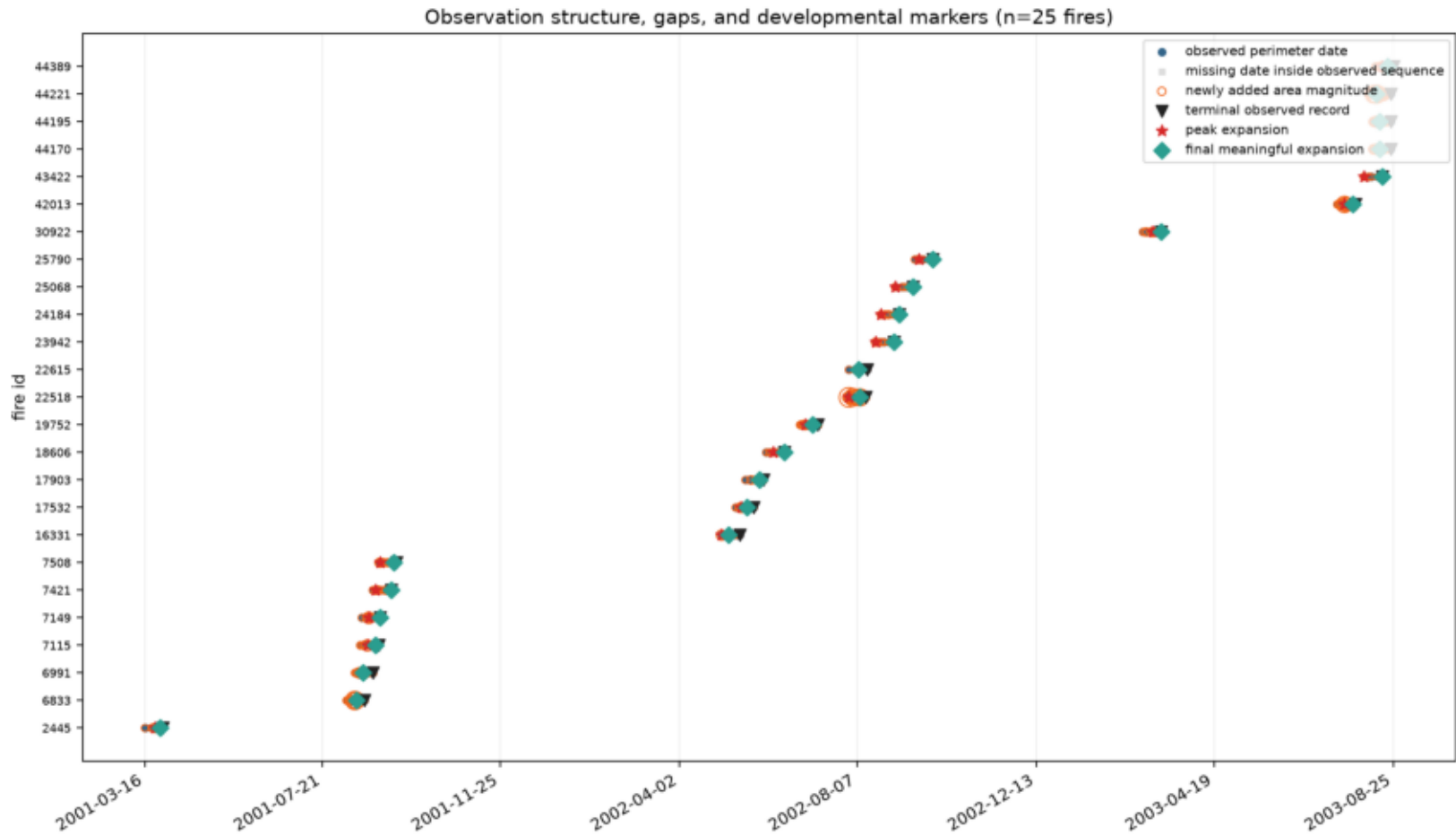
Strongest current manuscript direction: geometric developmental objects, event clocks, and continuous developmental axes.

Secondary direction: geometry-environment transitions around expansion pulses, but this needs larger sample size and better climate support.

Task-specific QC replaces terminal-cessation filtering

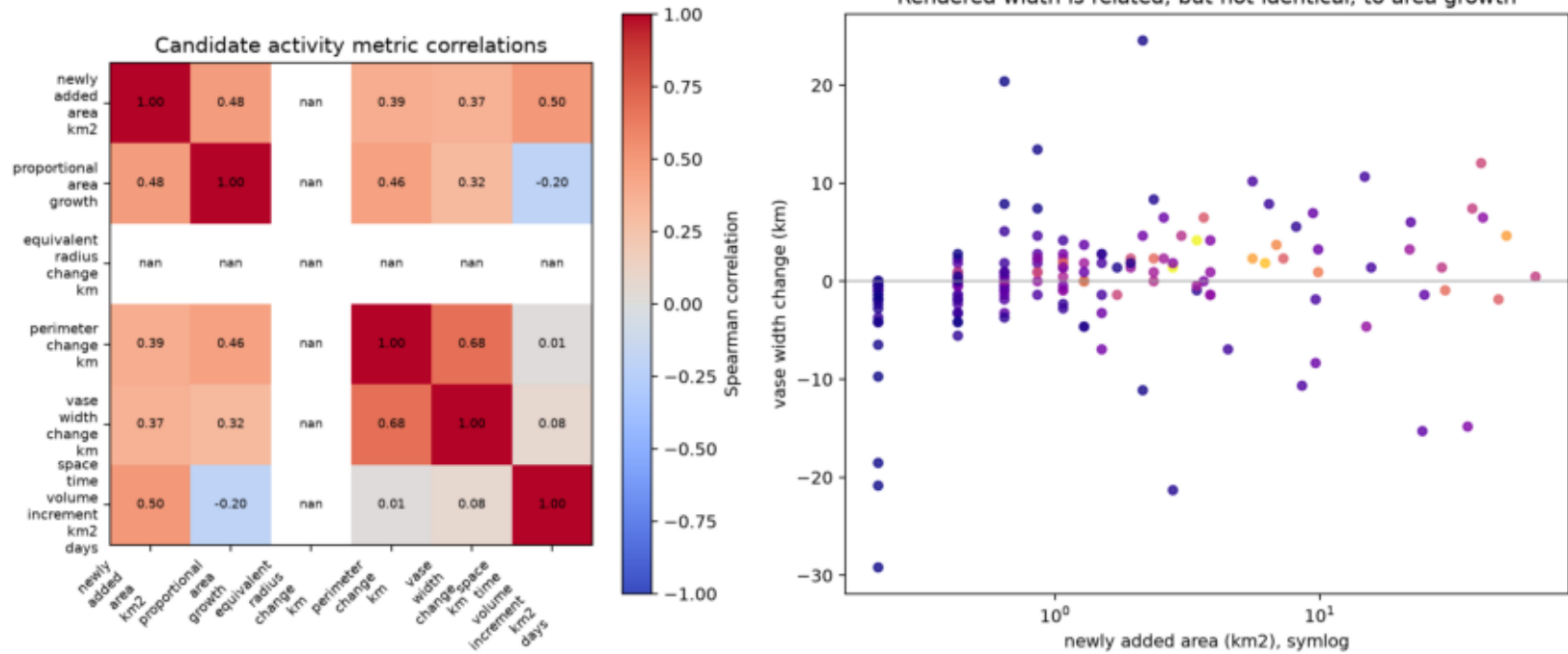
fire_id	observation count	observed_sequence_duration day	max_gap_days	expansion_pulse_count	usable_for_morphology	usable_for_pulse_detection	usable_for_lag_analysis	usable_for_terminal_observation	usable_for_clustering	usable_for_climate_transition
2445	8	13	5	2	True	True	False	False	True	True
6833	13	13	2	1	True	True	True	True	True	True
6991	12	13	3	2	True	True	True	False	True	True
7115	11	13	2	2	True	True	True	True	True	True
7149	9	13	3	2	True	True	True	False	True	True
7421	8	13	3	4	True	True	False	False	True	True
7508	8	13	3	2	True	True	False	False	True	True
16331	10	13	3	3	True	True	False	False	True	True
17532	7	13	4	1	True	True	True	False	True	True
17903	5	13	4	2	True	True	False	False	True	True
18606	7	13	4	5	True	True	True	False	True	True
19752	11	13	2	4	True	True	True	True	True	True
22518	14	13	1	4	True	True	False	True	True	True
22615	6	13	5	1	True	True	False	False	True	True
23942	8	13	4	6	True	True	False	False	True	True
24184	7	13	3	4	True	True	False	False	True	True
25068	5	13	5	4	True	True	False	False	True	True
25790	6	13	5	3	True	True	False	False	True	True
30922	9	13	4	4	True	True	True	False	True	True
42013	13	13	2	3	True	True	True	True	True	True
43422	6	13	5	4	True	True	False	False	True	True
44170	12	13	2	3	True	True	True	True	True	True
44195	10	13	3	2	True	True	True	False	True	True
44221	13	13	2	3	True	True	True	True	True	True
44389	11	13	2	2	True	True	True	True	True	True

Section 1. Observational structure and limitations



Observed perimeter dates, missing dates inside observed sequences, pulse markers, final meaningful expansion, and terminal observed records.

Section 2. Defining the developmental object

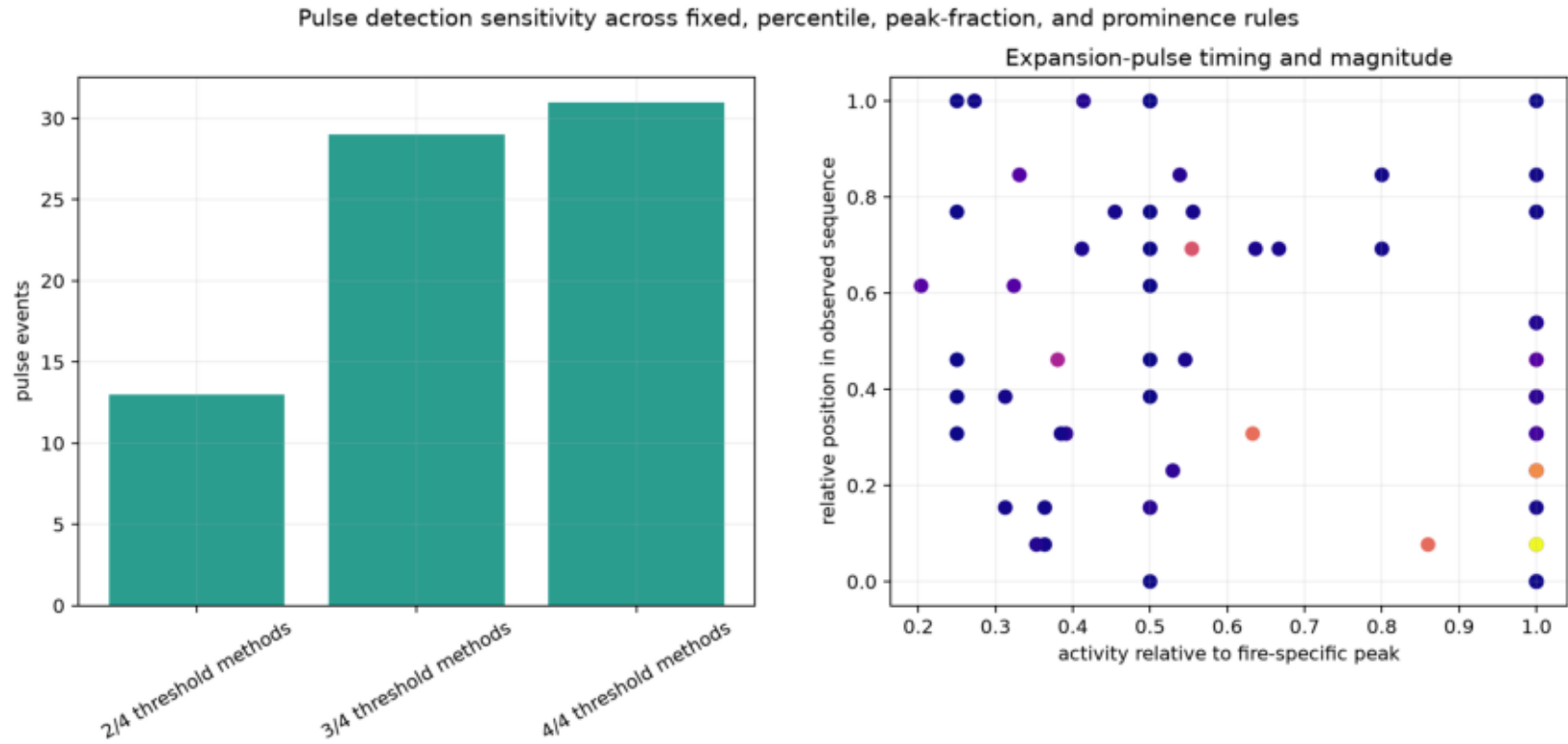


Vase width is max projected bounding-box side in km; it is compared against, not assumed equivalent to, newly added area.

Section 4. Event detection rules and detected events

fire_id	event_type	observation_index	calendar_date	activity_magnitude	activity_relative to peak	confidence_or_robustness
2445	first observed perimeter	0	2001-03-16 00:00:00	0.215	0.111	observed
2445	expansion pulse	2	2001-03-23 00:00:00	1.93	1	4/4 threshold methods
2445	peak expansion	2	2001-03-23 00:00:00	1.93	1	observed maximum
2445	expansion pulse	4	2001-03-26 00:00:00	1.07	0.556	geometric threshold-defined
2445	final meaningful expansion into	5	2001-03-27 00:00:00	0.644	0.333	4/4 threshold methods
2445	geometric	6	2001-03-28 00:00:00	0.215	0.111	final active rule-based
2445	terminal observed record	7	2001-03-29 00:00:00	0.215	0.111	data-based observed
6833	first observed perimeter	0	2001-08-07 00:00:00	0.215	0.00422	ending; not physical
6833	expansion pulse	5	2001-08-13 00:00:00	50.9	1	observed
6833	peak expansion	5	2001-08-13 00:00:00	50.9	1	4/4 threshold methods
6833	final meaningful expansion into	6	2001-08-14 00:00:00	29	0.57	observed maximum
6833	geometric	7	2001-08-15 00:00:00	8.59	0.169	geometric threshold-defined
6833	terminal observed record	12	2001-08-20 00:00:00	0.215	0.00422	final active rule-based
6991	first observed perimeter	0	2001-08-13 00:00:00	0.215	0.0217	data-based observed
6991	expansion pulse	4	2001-08-17 00:00:00	3.86	0.391	ending; not physical
6991	expansion pulse	6	2001-08-19 00:00:00	9.87	1	observed
6991	final meaningful expansion	6	2001-08-19 00:00:00	9.87	1	3/4 threshold methods
6991	peak expansion transition into	6	2001-08-19 00:00:00	9.87	1	4/4 threshold methods
6991	geometric	7	2001-08-20 00:00:00	1.07	0.109	threshold-defined
6991	terminal observed record	11	2001-08-26 00:00:00	0.429	0.0435	final active observed maximum
7115	first observed perimeter	0	2001-08-17 00:00:00	0.429	0.0769	geometric rule-based
7115	expansion pulse	3	2001-08-22 00:00:00	5.58	1	data-based observed
7115	peak expansion	3	2001-08-22 00:00:00	5.58	1	activity
7115	expansion pulse	8	2001-08-28 00:00:00	3.01	0.538	ending; not physical
7115	final meaningful expansion into	8	2001-08-28 00:00:00	3.01	0.538	observed
7115	geometric	9	2001-08-29 00:00:00	0.429	0.0769	4/4 threshold methods
7115	terminal observed record	10	2001-08-30 00:00:00	0.215	0.0385	threshold-defined
7149	first observed perimeter	0	2001-08-18 00:00:00	0.215	0.0345	final active rule-based

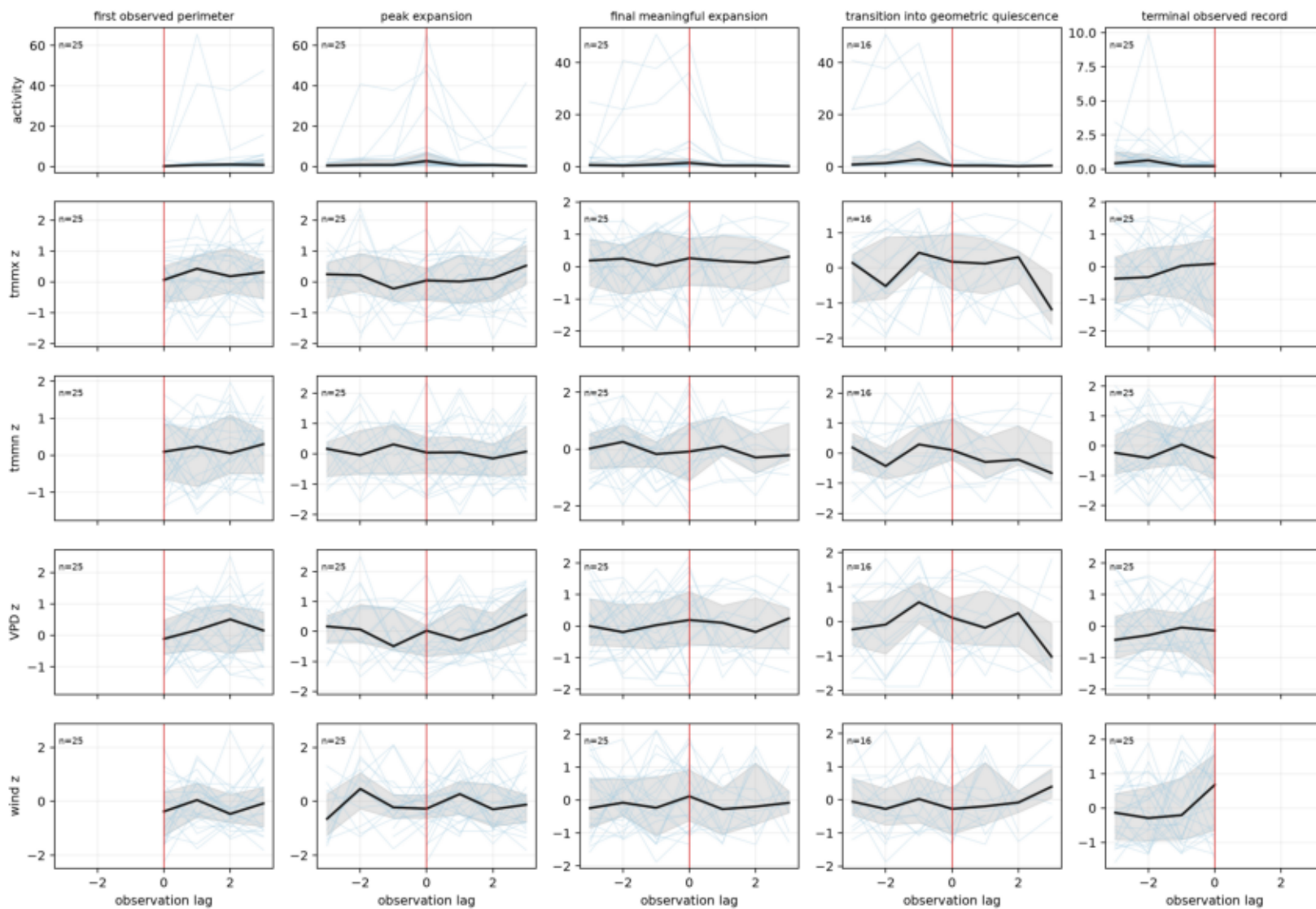
Section 4. Event detection sensitivity



Expansion activity is detected from geometry only using multiple transparent thresholds.

Section 5. Alignment comparison

Which event provides the most coherent fire-centered clock?

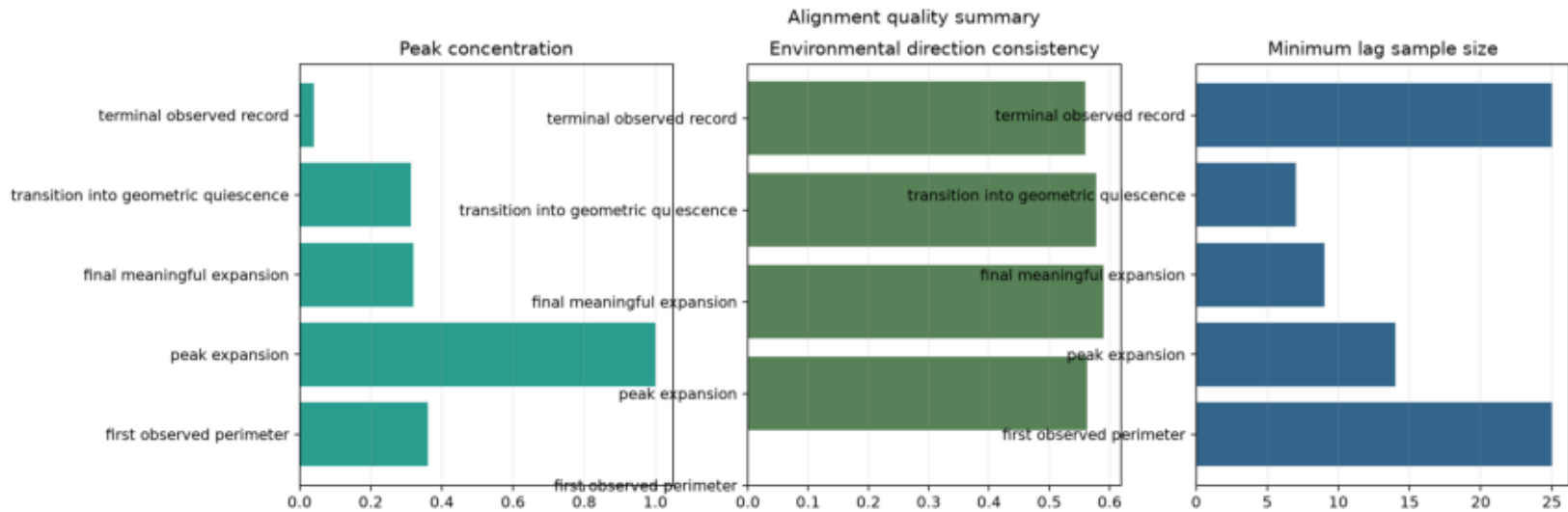


Observed points and event-centered lags are shown; interpolated relative-age views are secondary diagnostics only.

Section 5. Alignment quality ranking

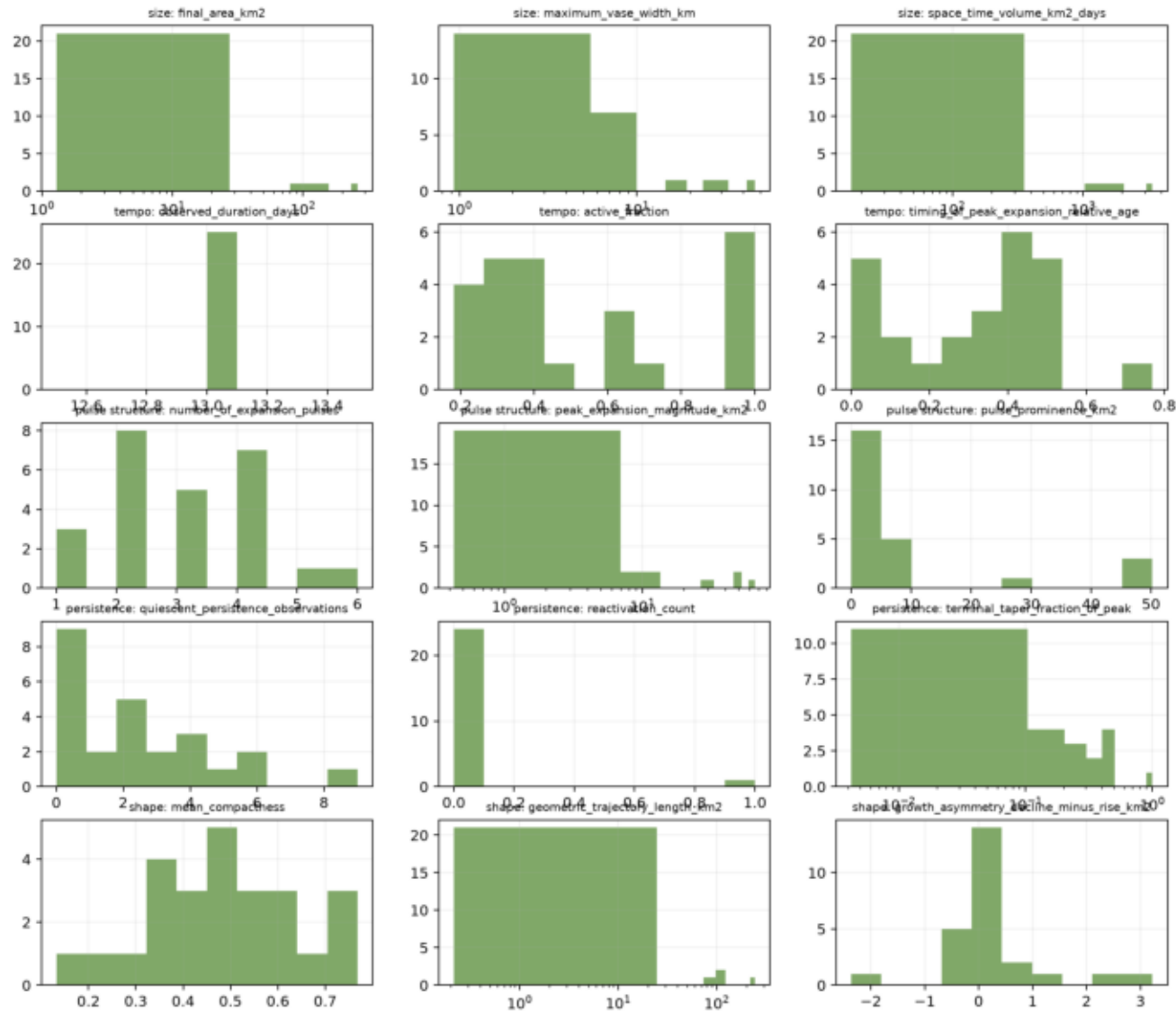
alignment_clock	n_fires	geometric_peak_concentration_within_1_obs	mean_among_fire_activity_variance	environmental_direction_consistency	minimum_lag_sample_size	question_answered	alignment_rank_for_geometry
first observed perimeter	25	0.36	92.9	nan	25	Which event provides the most coherent fire-centered clock?	2
peak expansion	25	1	90.3	0.562	14	Which event provides the most coherent fire-centered clock?	1
final meaningful expansion	25	0.32	63	0.59	9	Which event provides the most coherent fire-centered clock?	3
transition into geometric quiescence	16	0.312	84.5	0.578	7	Which event provides the most coherent fire-centered clock?	4
terminal observed record	25	0.04	1.24	0.56	25	Which event provides the most coherent fire-centered clock?	5

Section 5. Alignment quality visuals



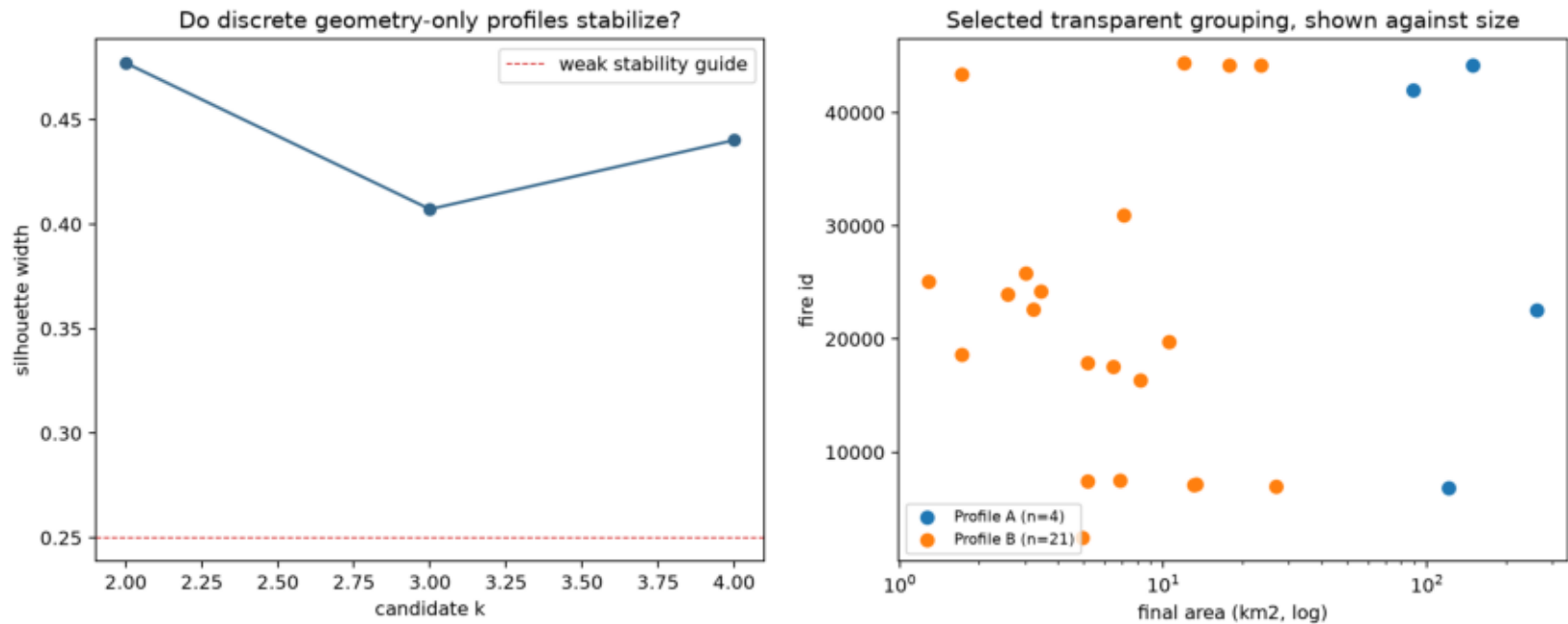
Section 6. Geometric life-history traits

Geometric life-history traits organized by scientific role



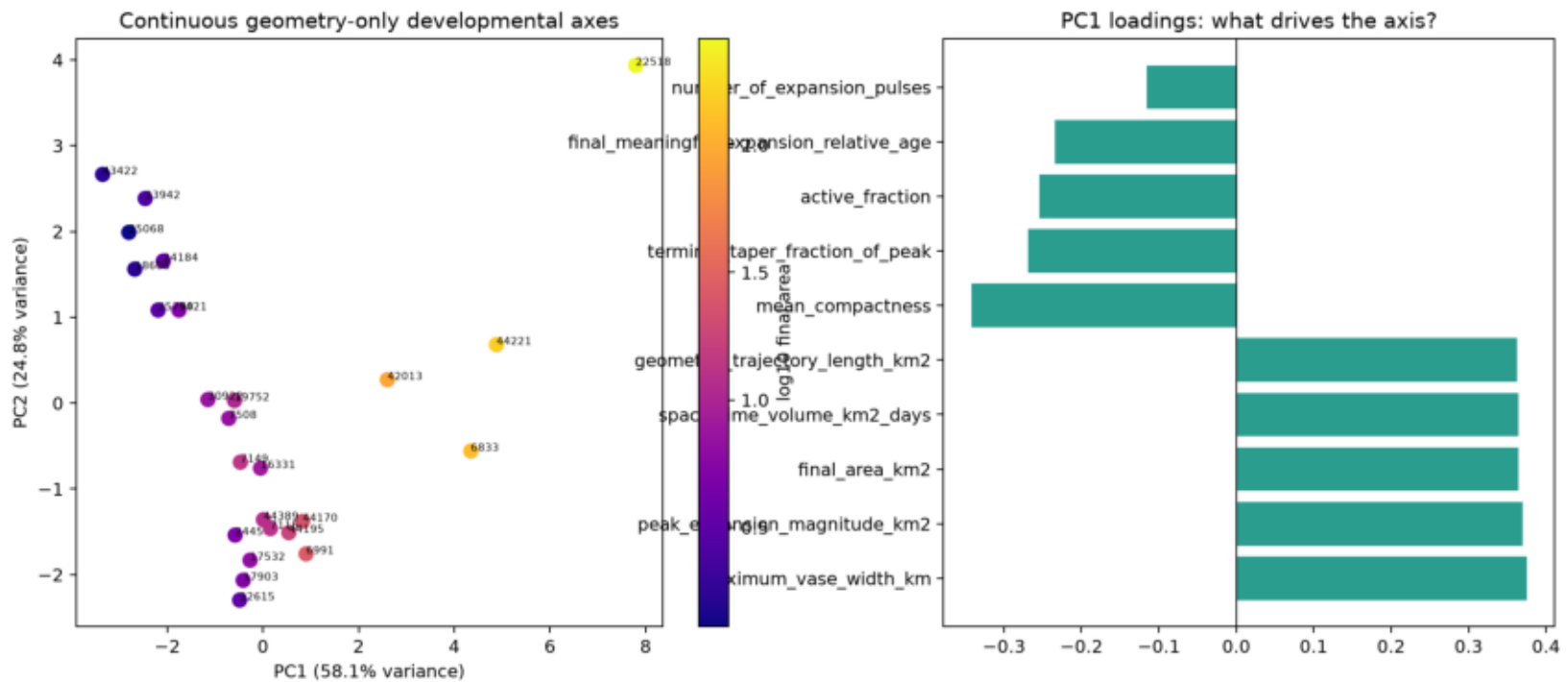
Traits are grouped as size, tempo, pulse structure, persistence, and shape; skewed axes use log scales where appropriate.

Section 7. Geometry-only developmental typology



Clusters use geometry only. Weak silhouette values mean continuous axes may be more appropriate than forced types.

Section 8. Continuous developmental axes



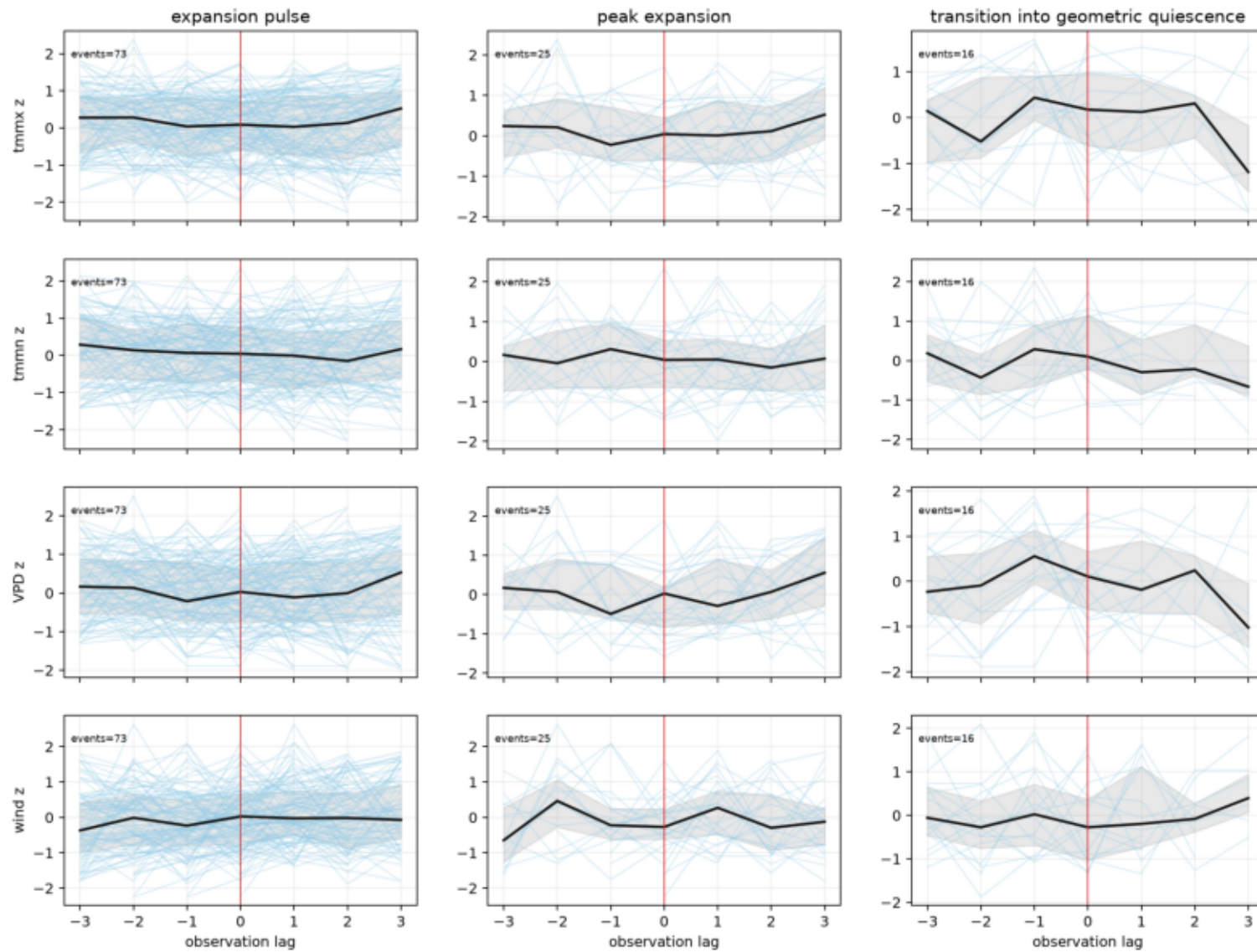
PCA scores are accompanied by loadings so interpretation remains tied to measurable traits.

PCA loadings for continuous geometry axes

variable	PC1_loading	PC2_loading
final_area_km2	0.365	0.225
observed_duration_days	1.11e-16	-2.78e-17
active_fraction	-0.254	0.421
peak_expansion_magnitude_km2	0.37	0.168
timing_of_peak_expansion_relative_age	0.0569	-0.428
final_meaningful_expansion_relative_age	-0.234	0.291
number_of_expansion_pulses	-0.115	0.483
terminal_taper_fraction_of_peak	-0.269	0.314
geometric_trajectory_length_km2	0.362	0.227
space_time_volume_km2_days	0.365	0.225
maximum_vase_width_km	0.375	0.175
mean_compactness	-0.342	0.101

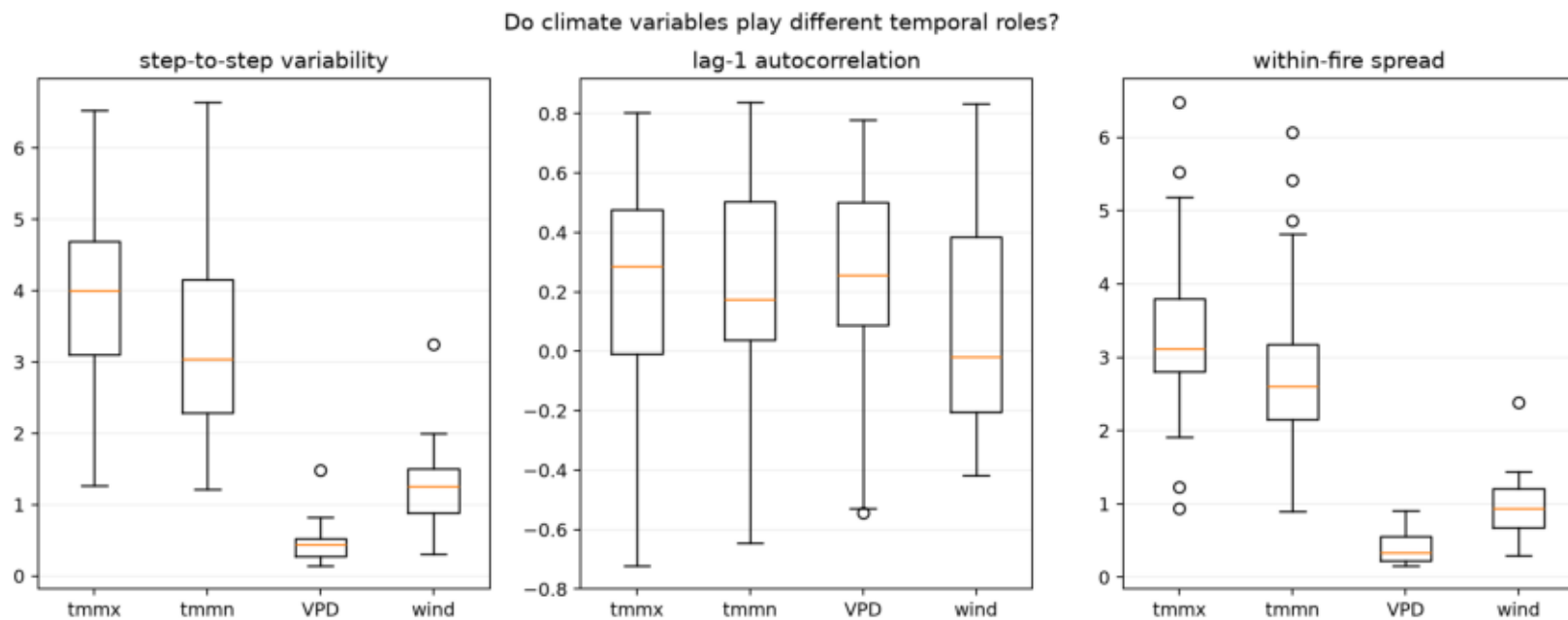
Section 9. Environment around expansion events

Environment around geometry-defined developmental events



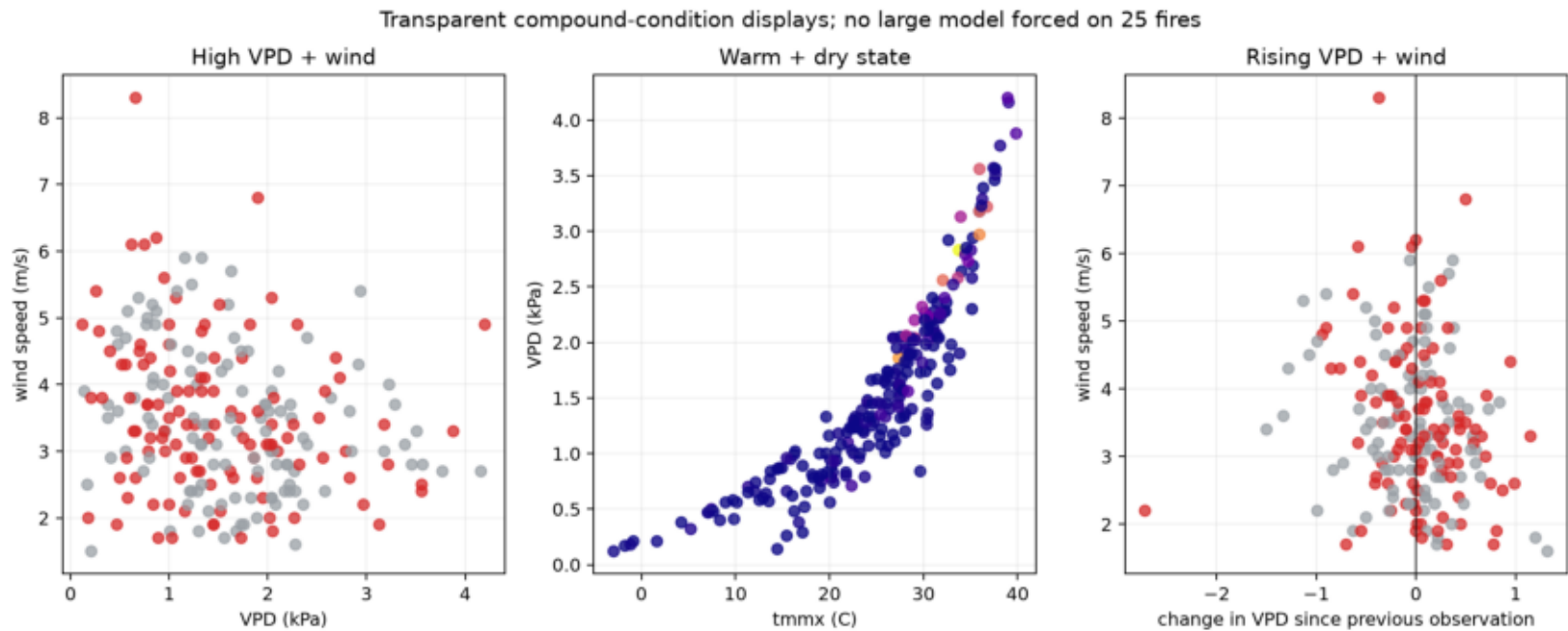
Climate describes geometry-defined events; it does not define them.

Section 10. Environmental roles



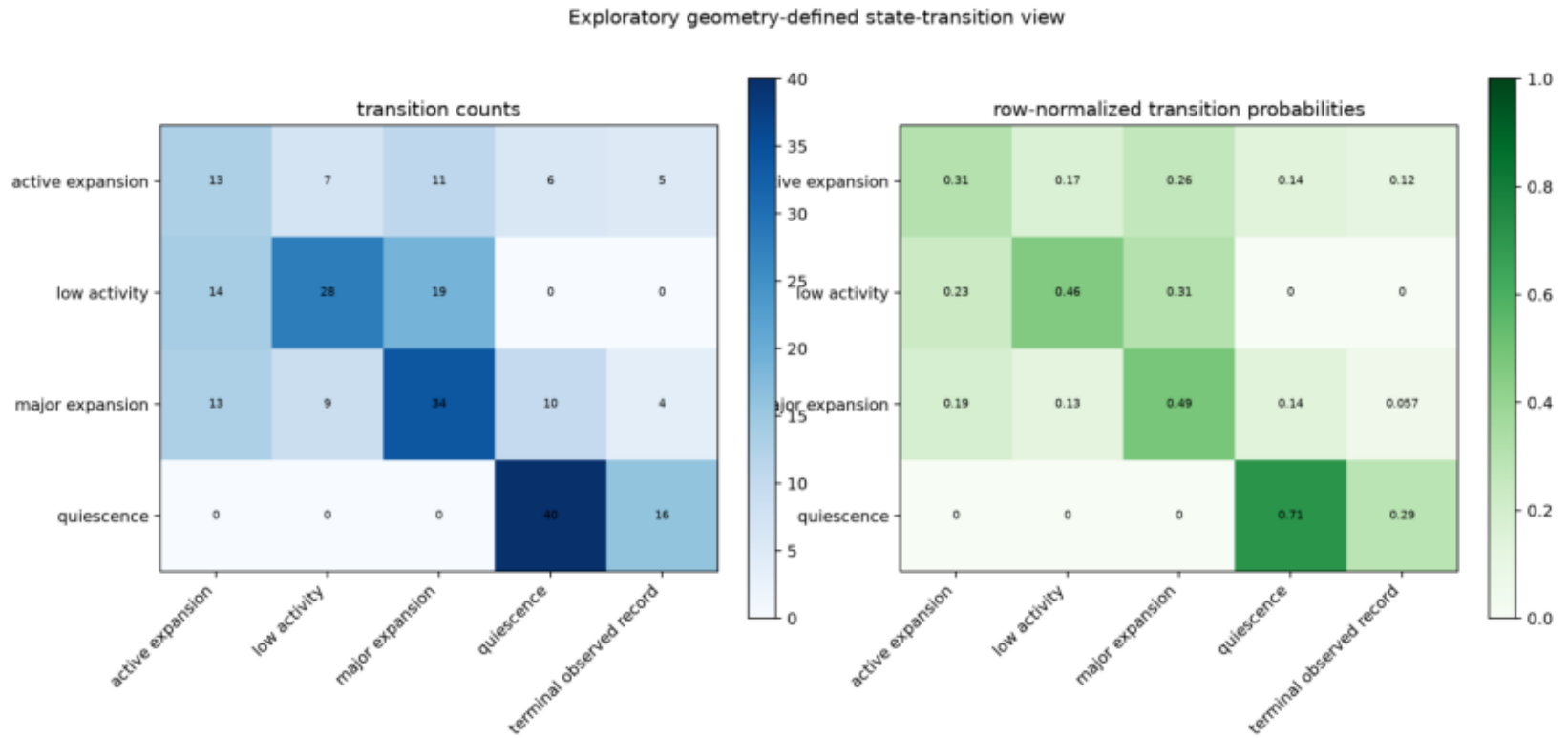
Daily maximum/minimum temperature, VPD, and scalar wind speed are compared as different temporal signals.

Section 11. Interactions and compound conditions



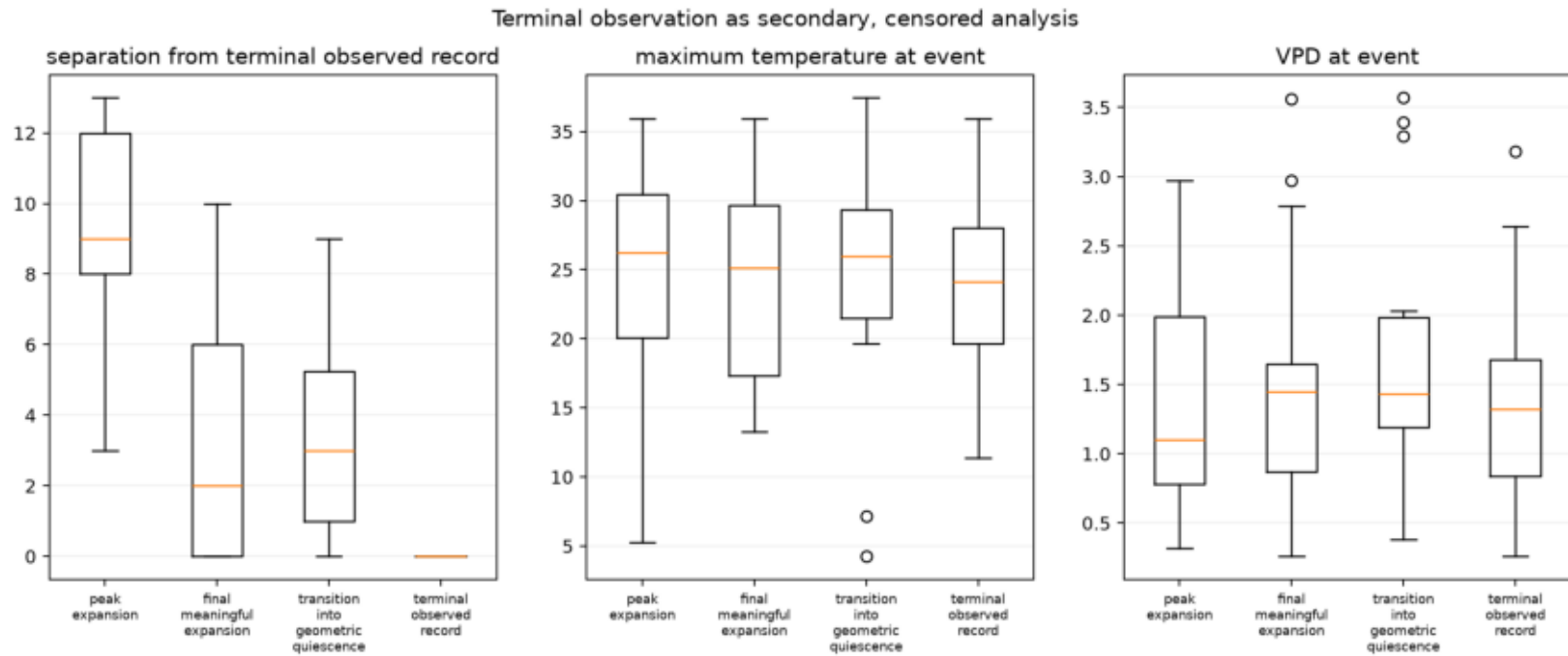
Transparent two-variable displays are used instead of over-parameterized models.

Section 12. State-transition view



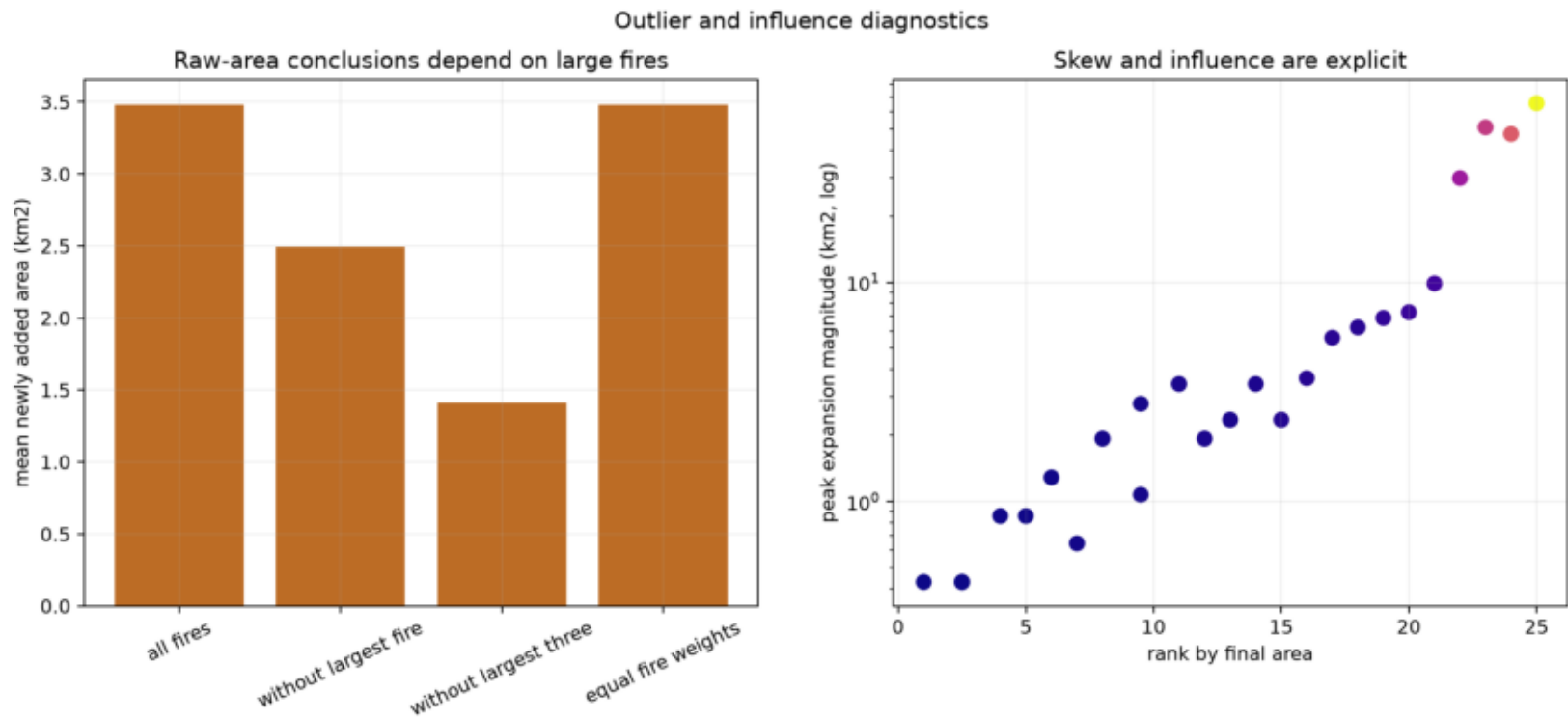
States are exploratory geometry bins, not verified biological states.

Section 13. Terminal observation as secondary analysis



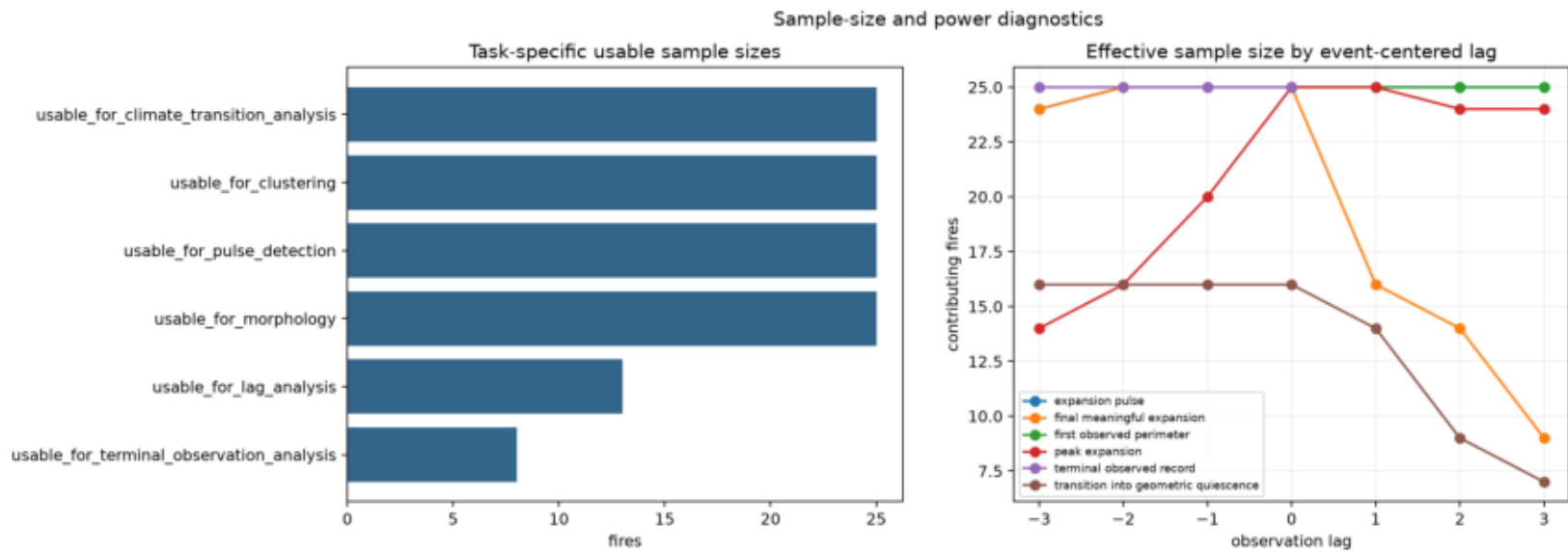
The terminal observed record is compared to peak, final meaningful expansion, and quiescence onset.

Section 14. Outlier and influence analysis



Large fires are shown explicitly; they are not silently removed.

Section 15. Sample-size and power diagnostics



Every major comparison reports fires, events, or lag contributions.

Section 16. Manuscript decision matrix: overview

story	current_evidence_rating	manuscript_potential	sample_size	robustness	strongest_figure
Wildfires converge on a common terminal climate.	unsupported with current data	low now	25	low	13_terminal_secondary_analysis
Wildfires share a common transition out of their final expansion pulse.	suggestive	moderate	25	medium	04_alignment_comparison_grid
Wildfires exhibit recurring geometric developmental trajectories.	suggestive	moderate	25	medium-low	07_geometry_clustering
Wildfire development lies along continuous geometric axes rather than discrete types.	moderate descriptive evidence	high	25	medium	08_continuous_axes_pca
Expansion pulses are associated with specific environmental states or changes.	suggestive	moderate after expansion	73	low-medium	09_event_centered_climate
Different climate variables play distinct temporal roles in wildfire development.	suggestive	moderate	25	low-medium	10_environmental_roles
Observation gaps prevent reliable inference about physical fire cessation.	strong descriptive evidence	high as limitation/design point	25	high	01_observational_structure_raster

Section 16. Manuscript decision matrix: evidence details

story	supporting_analyses	contradicting_analyses	dependence_on_definitions	additional_analysis_needed
Wildfires converge on a common terminal climate.	terminal secondary analysis	wide terminal tmmx/VPD/wind spread; censored terminal records	high	true cessation data; expanded sample; boundary climate
Wildfires share a common transition out of their final expansion pulse.	alignment comparison; final meaningful expansion events	gaps and varying quiescence persistence	medium	test pulse thresholds on larger sample
Wildfires exhibit recurring geometric developmental trajectories.	profiles; event table; clustering	cluster silhouette max 0.48	medium	stability with more fires
Wildfire development lies along continuous geometric axes rather than discrete types.	PCA loadings and scores	small sample size	medium	functional PCA and outlier sensitivity
Expansion pulses are associated with specific environmental states or changes.	event-centered climate extraction	daily time step, nearest-grid-cell climate, limited pulse count	high	larger sample; boundary climate; hourly wind optional
Different climate variables play distinct temporal roles in wildfire development.	environmental roles diagnostics	daily records smooth short wind events	medium	more events and subdaily wind if needed
Observation gaps prevent reliable inference about physical fire cessation.	observational raster; task-specific QC	none in current cache	low	independent cessation observations

Section 17. Recommended main figures

OPTION A: GEOMETRIC DEVELOPMENT PAPER

Figure 1: The vase as a fire developmental object - current data are sufficient for a descriptive figure.

Figure 2: Individual developmental profiles and detected events - current profiles are sufficient, but event rules should be sensitivity-tested on more fires.

Figure 3: Geometry-only continuous axes or stable trajectory groups - PCA axes are currently stronger than forced clusters.

Figure 4: Event-aligned expansion and quiescence transitions - current data are suggestive but limited by gaps.

Figure 5: Conceptual synthesis of wildfire developmental trajectories - appropriate as a hypothesis-generating synthesis.

OPTION B: GEOMETRY-ENVIRONMENT PAPER

Figure 1: The vase and aligned geometry-climate time series - current data are sufficient as an atlas figure.

Figure 2: Environmental conditions around major expansion pulses - suggestive only with current sample size.

Figure 3: Compound climate conditions and lagged response - needs more fires and boundary climate support.

Figure 4: Developmental states and transition probabilities - exploratory, not mechanistic.

Figure 5: Environmental trajectories across geometric development - promising but requires richer climate support.

Section 18. Final scientific interpretation - page 1

What the current data show

Observation: the cached sample contains daily fire perimeters, gaps, and nearest-grid-cell daily climate for 25 candidate fires.

Statistical result: expansion activity is uneven, with pulse-like growth, low-growth periods, and variable timing of final meaningful expansion.

Scientific inference: the current sample is consistent with fire development being organized around geometric events rather than a verified extinction endpoint.

Speculation: some fires may occupy recurring developmental pathways, but discrete groups are not yet stable enough to name mechanistically.

Section 18. Final scientific interpretation - page 2

What the current data do not yet establish

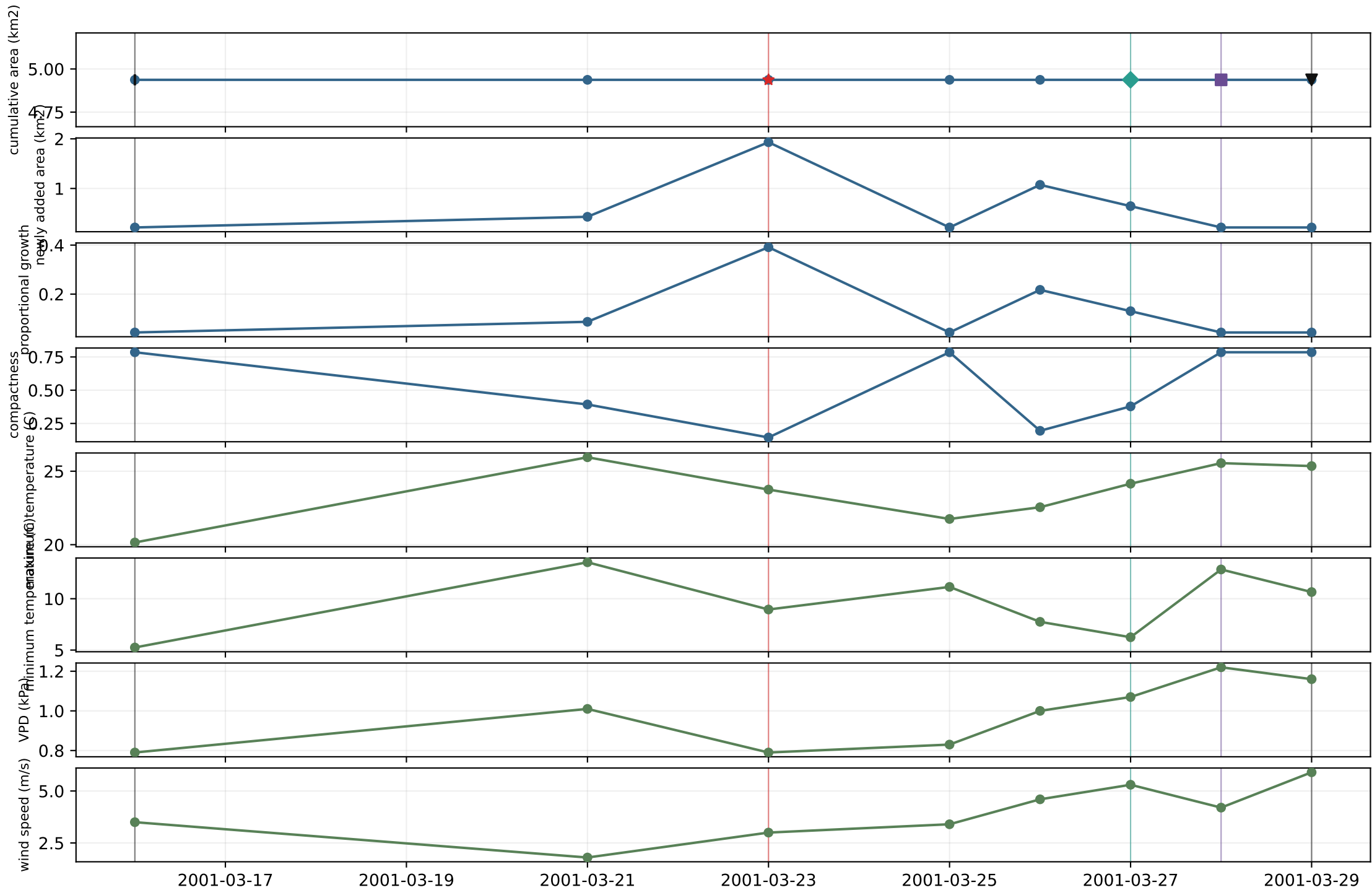
The current data do not distinguish terminal observed record from physical extinction.

The current data do not establish causal climate controls on expansion pulses.

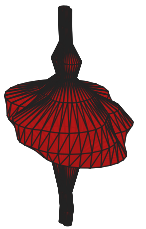
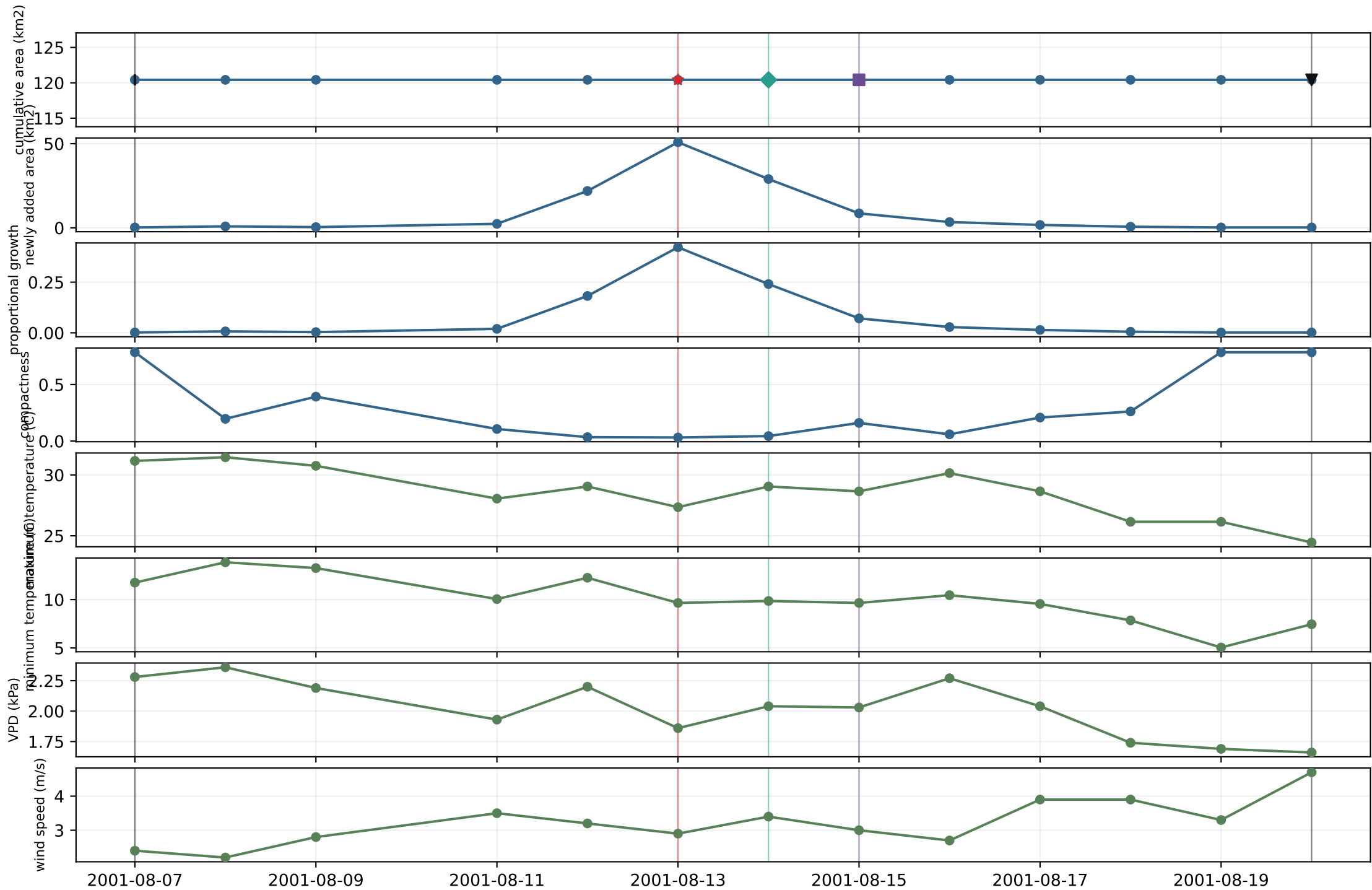
This pattern is sensitive to observation gaps, pulse thresholds, and large-fire influence.

This result is strongest as a scientific decision atlas for manuscript selection, not as a polished causal argument.

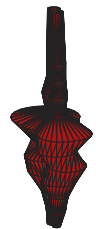
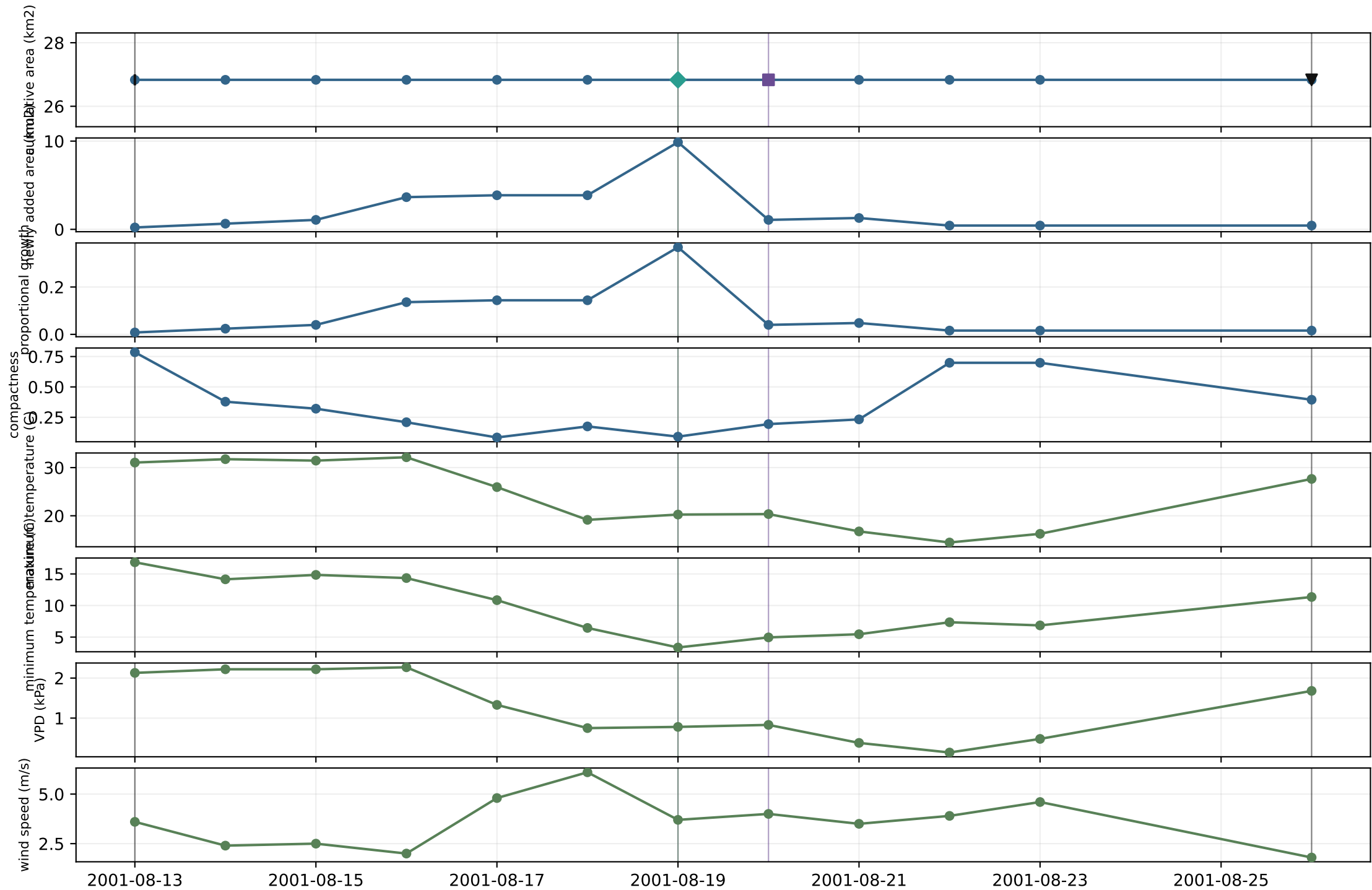
Fire 2445: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



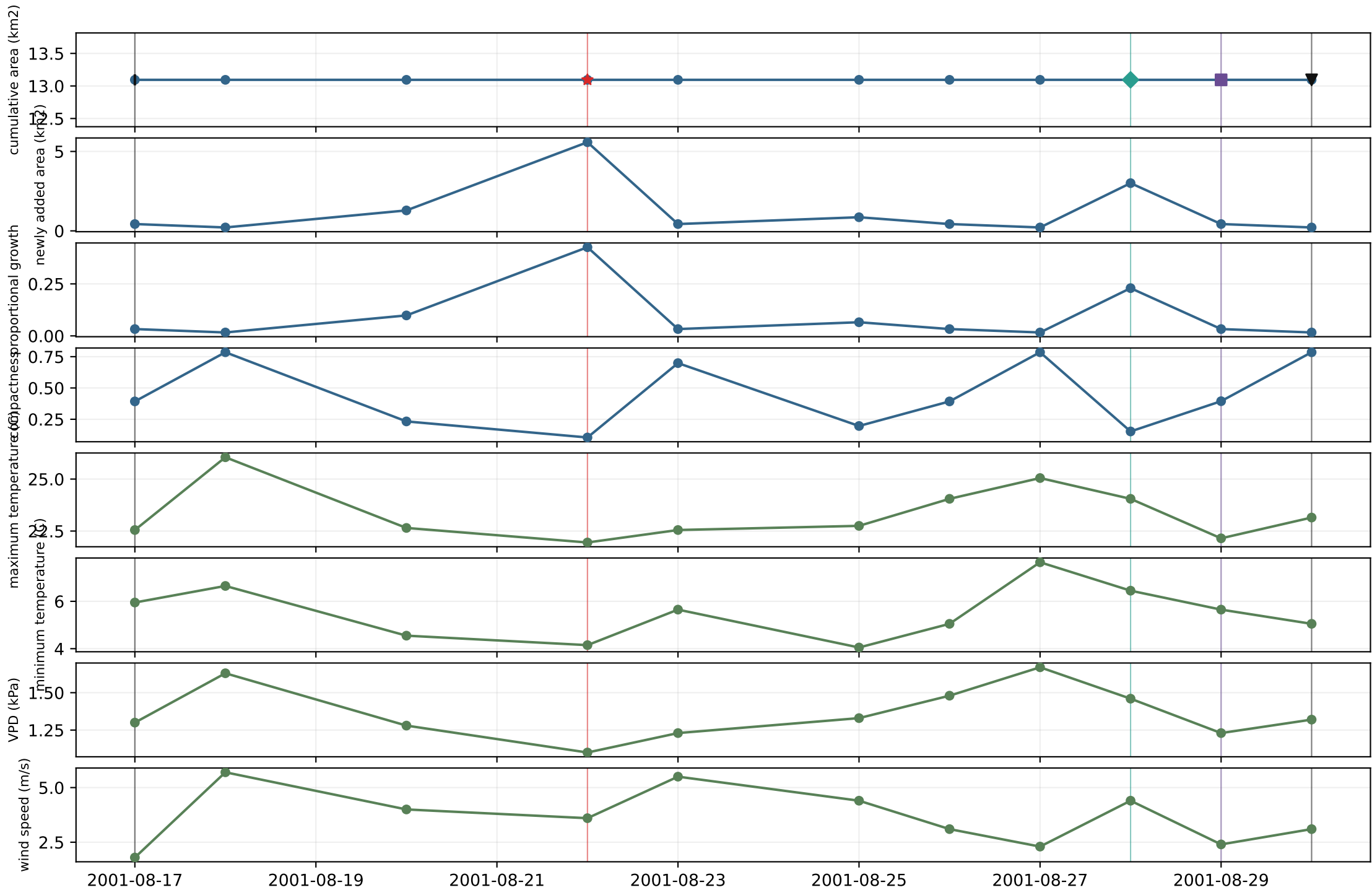
Fire 6833: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



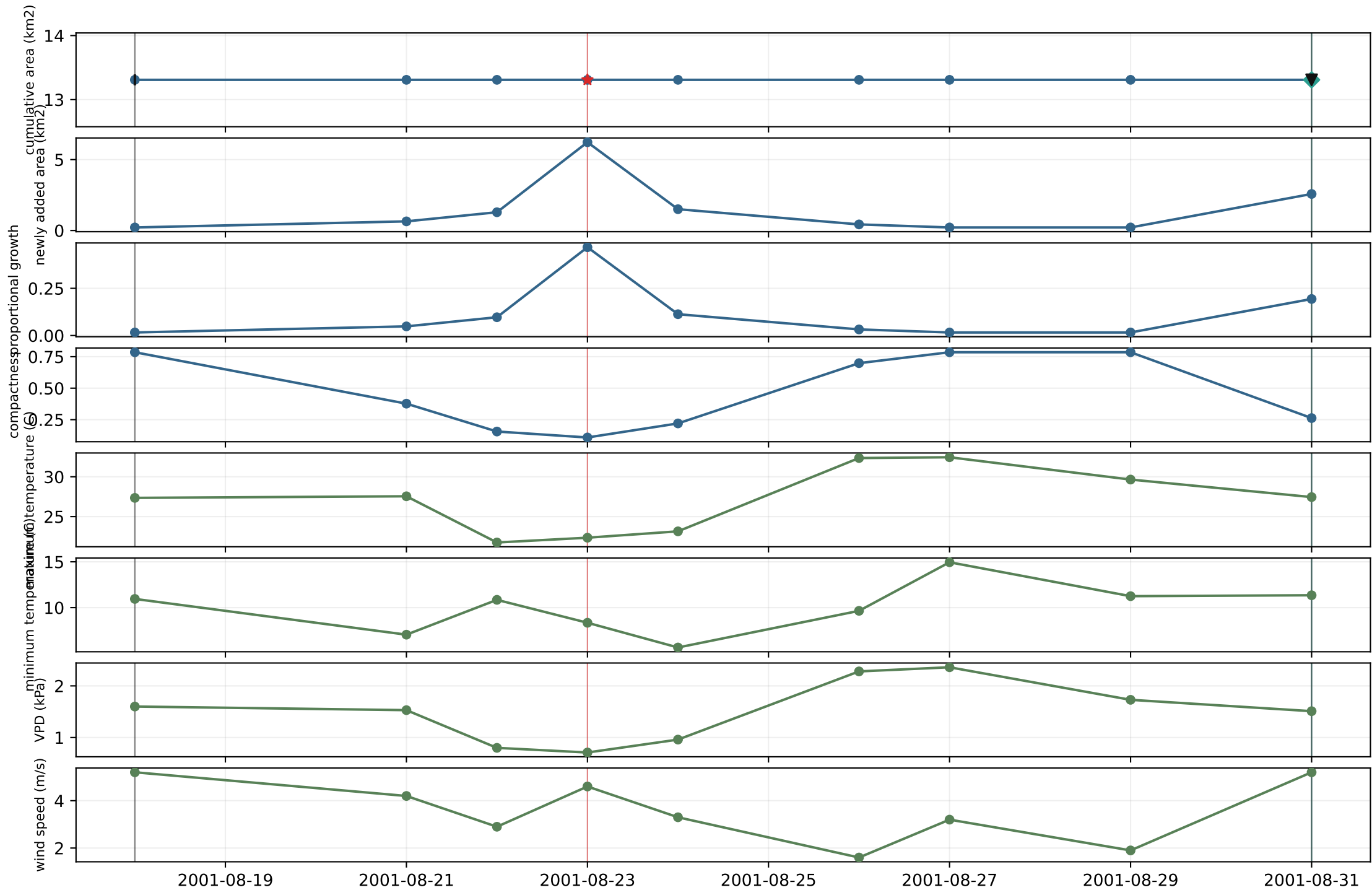
Fire 6991: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



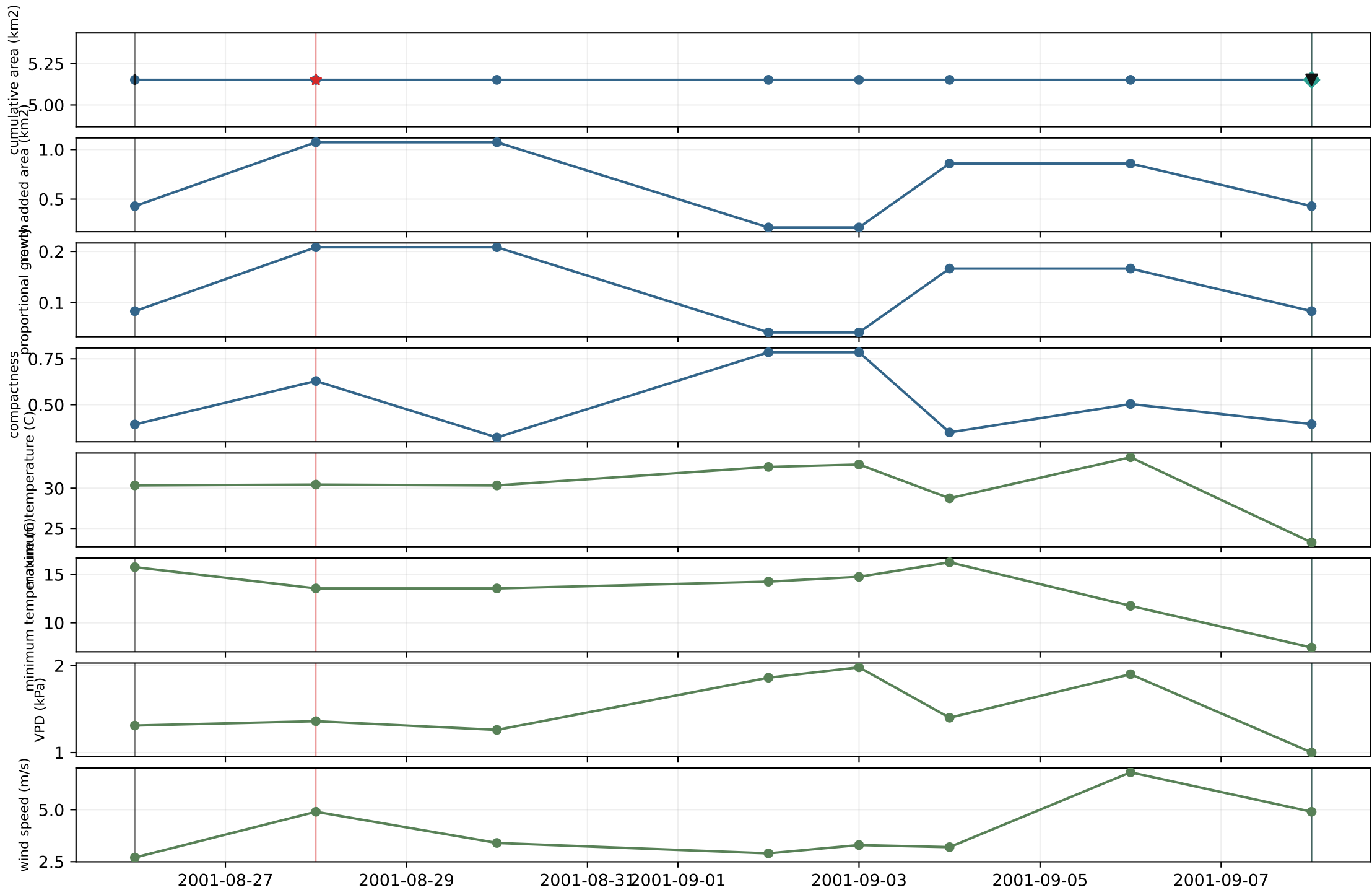
Fire 7115: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



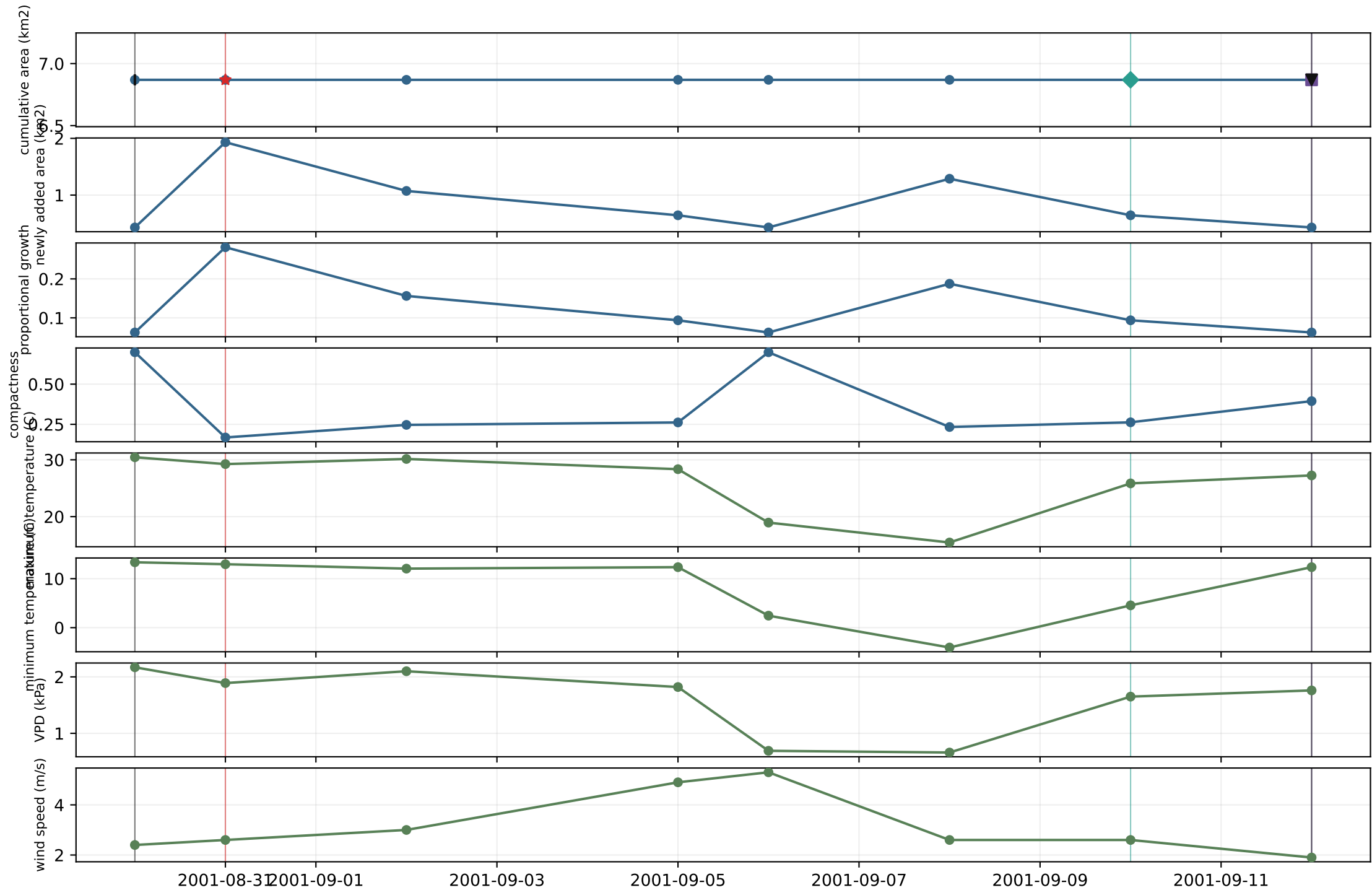
Fire 7149: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



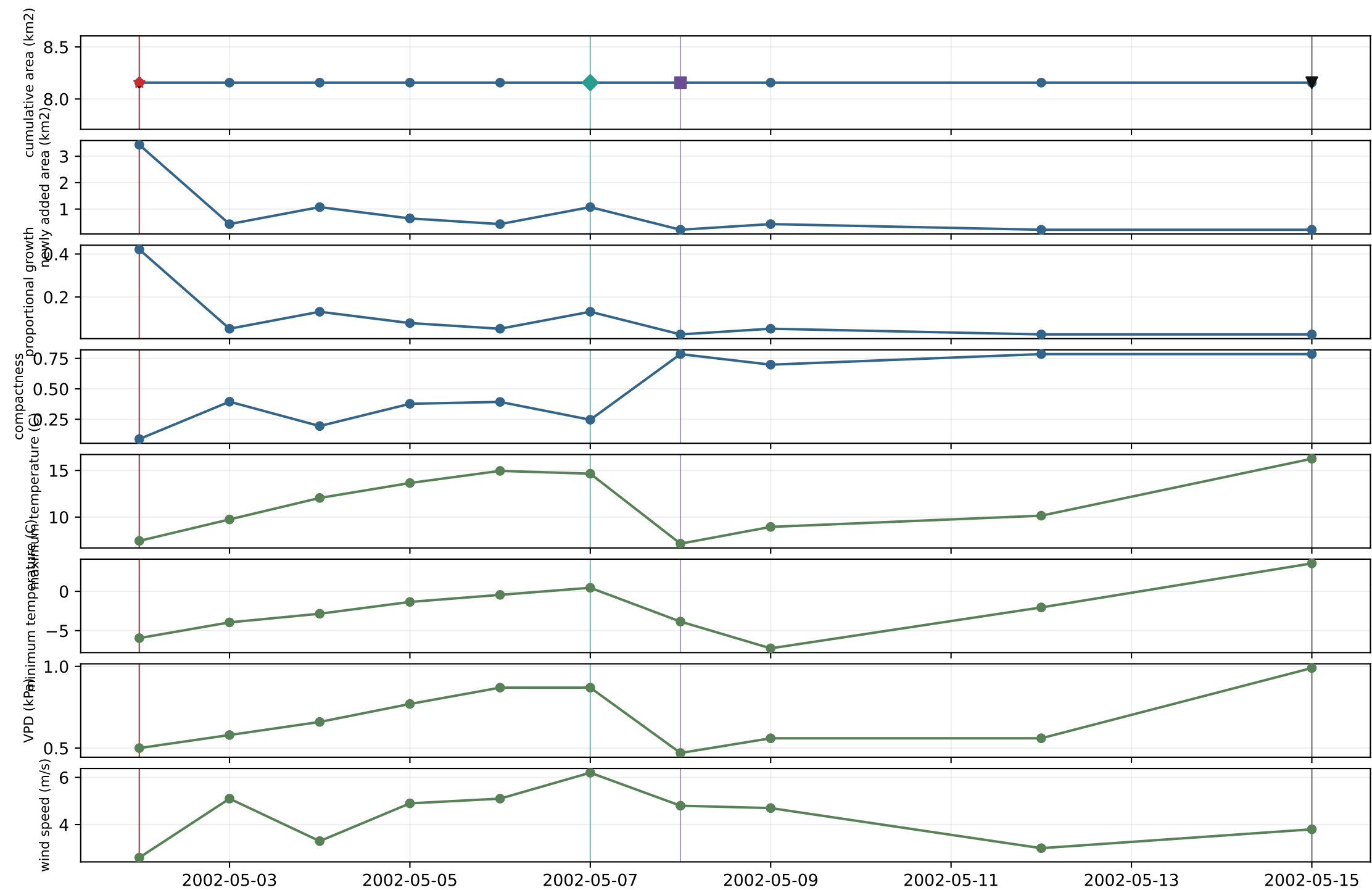
Fire 7421: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



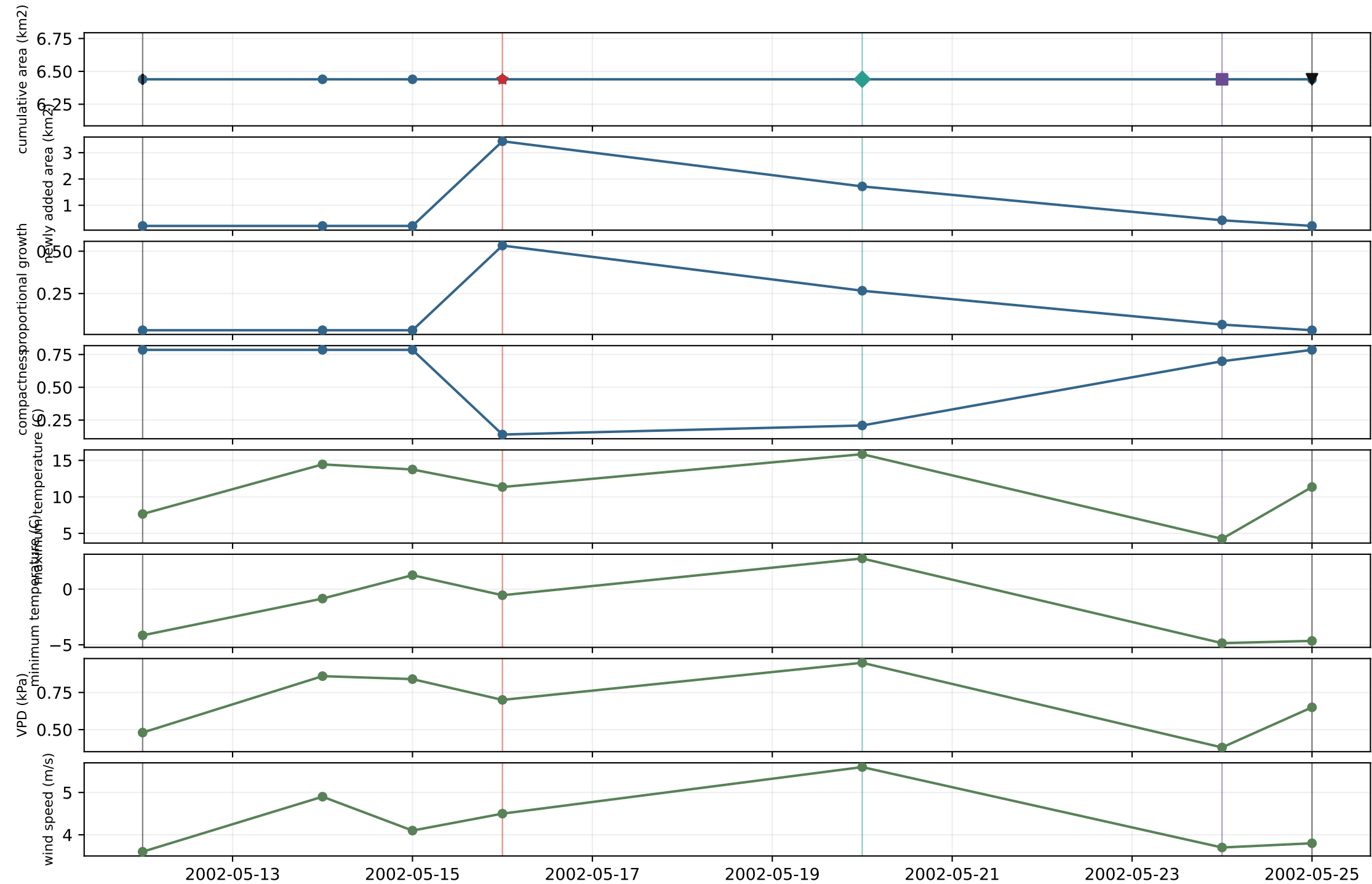
Fire 7508: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



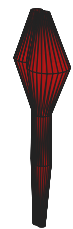
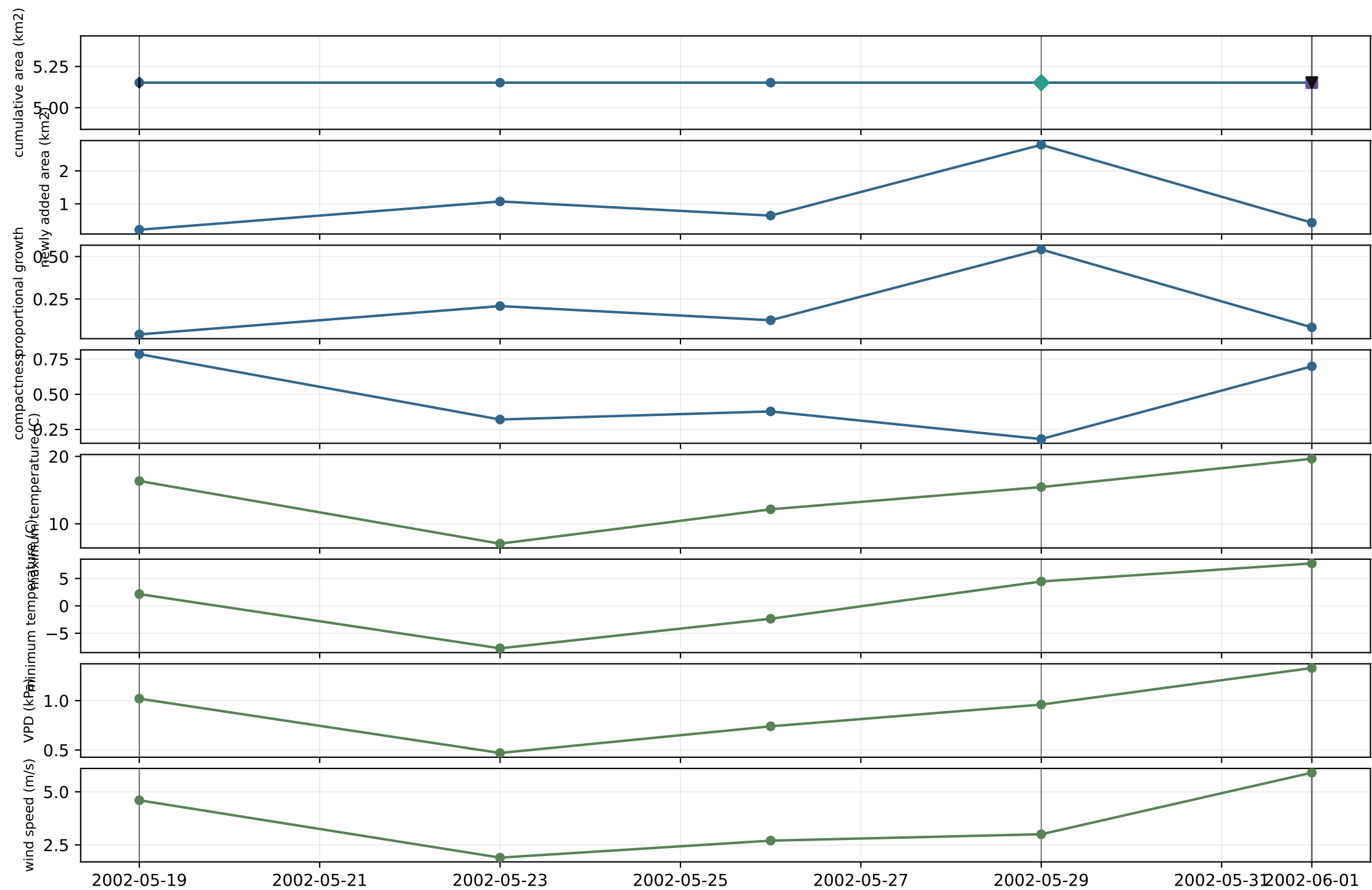
Fire 16331: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



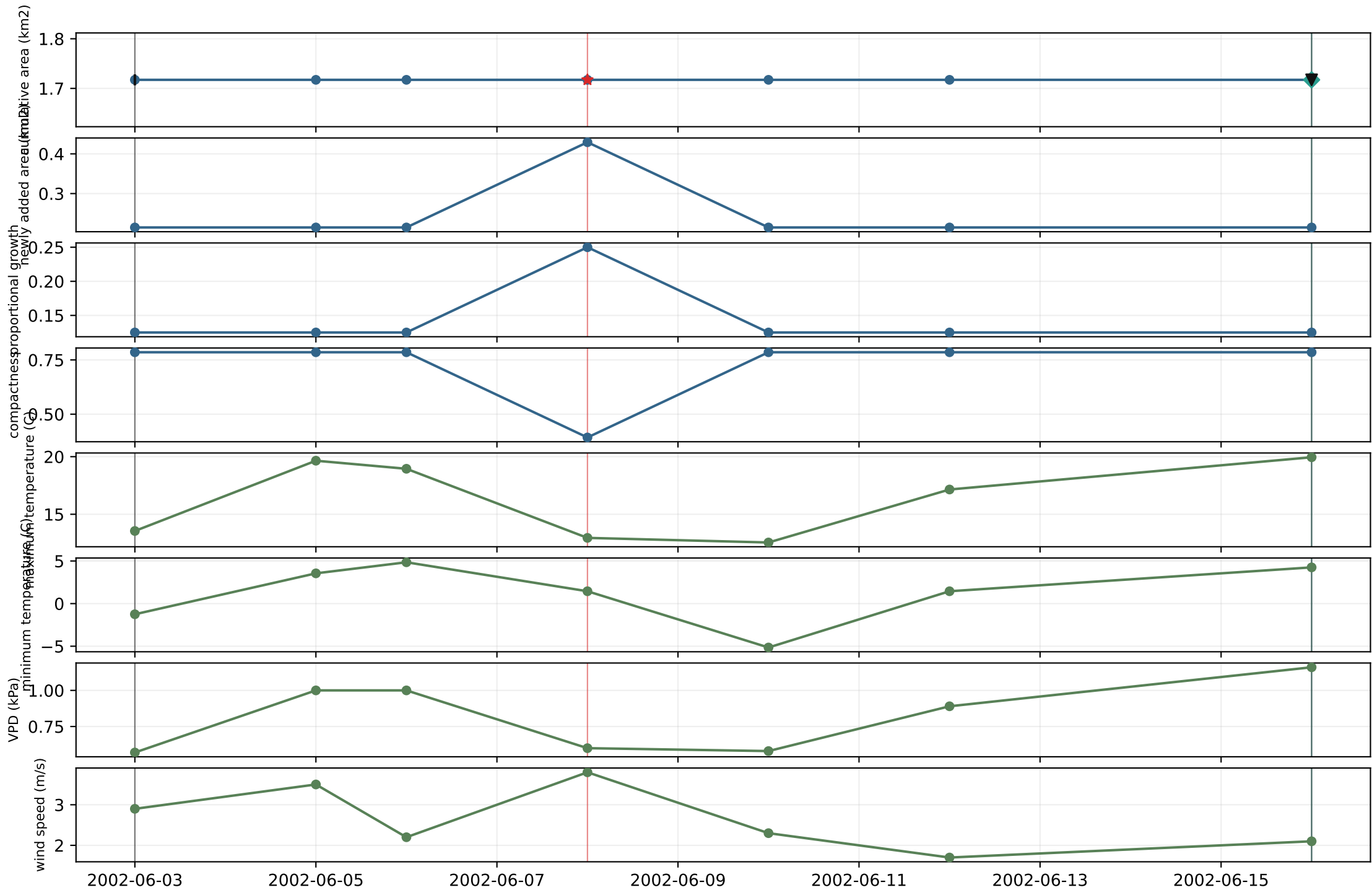
Fire 17532: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



Fire 17903: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction

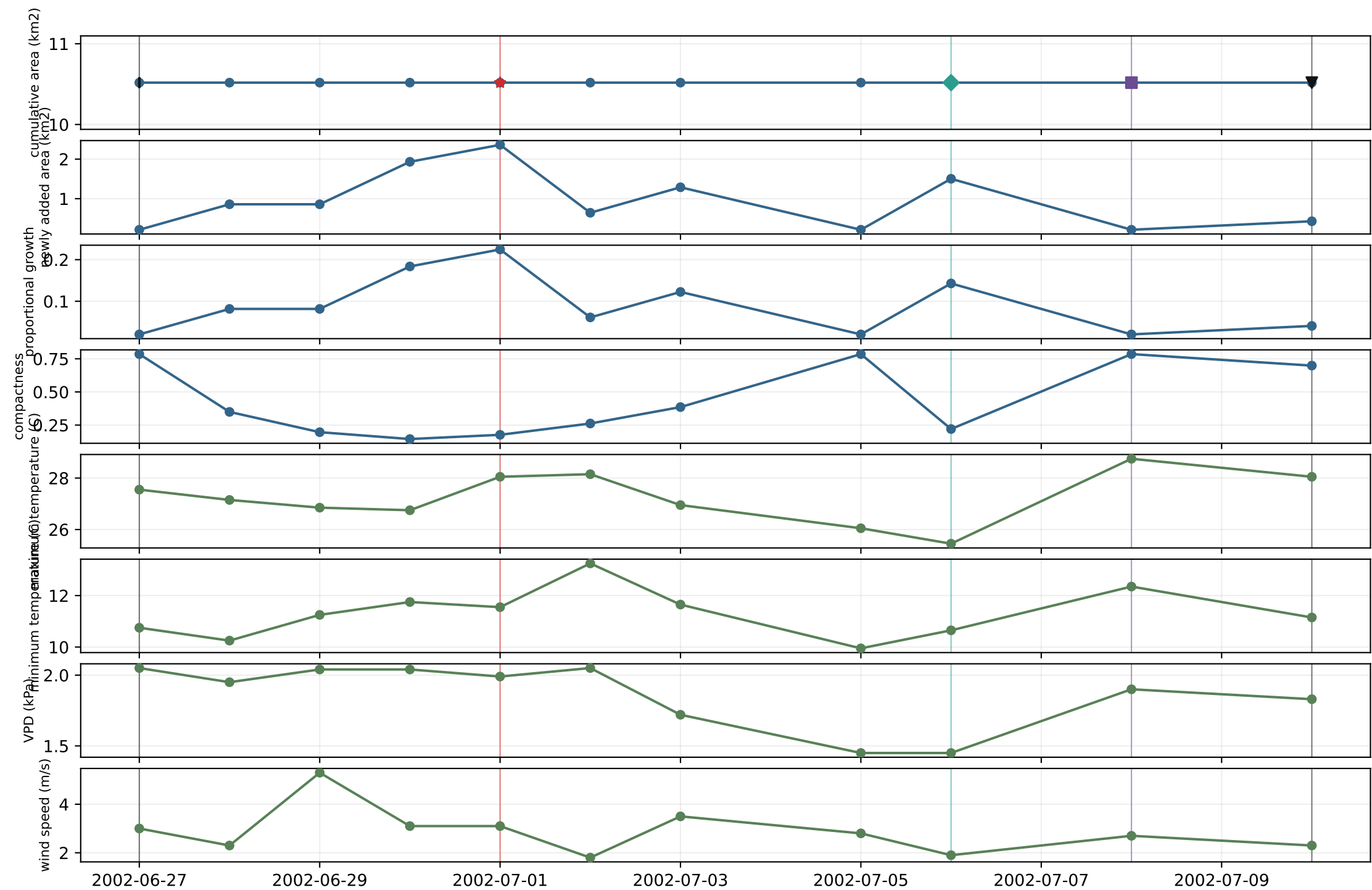


Fire 18606: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction

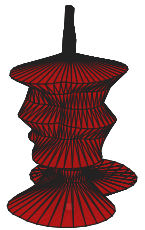
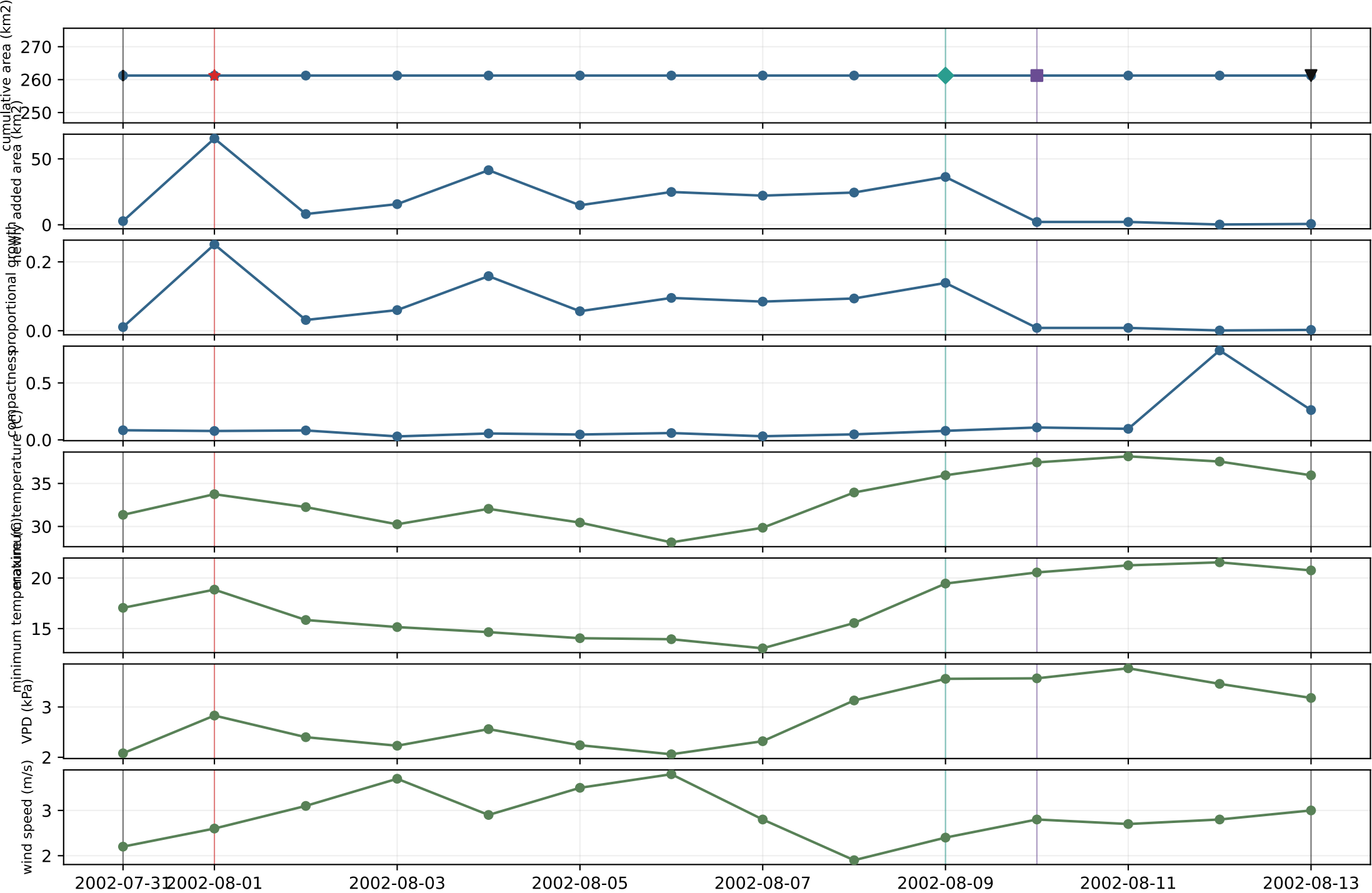


18606

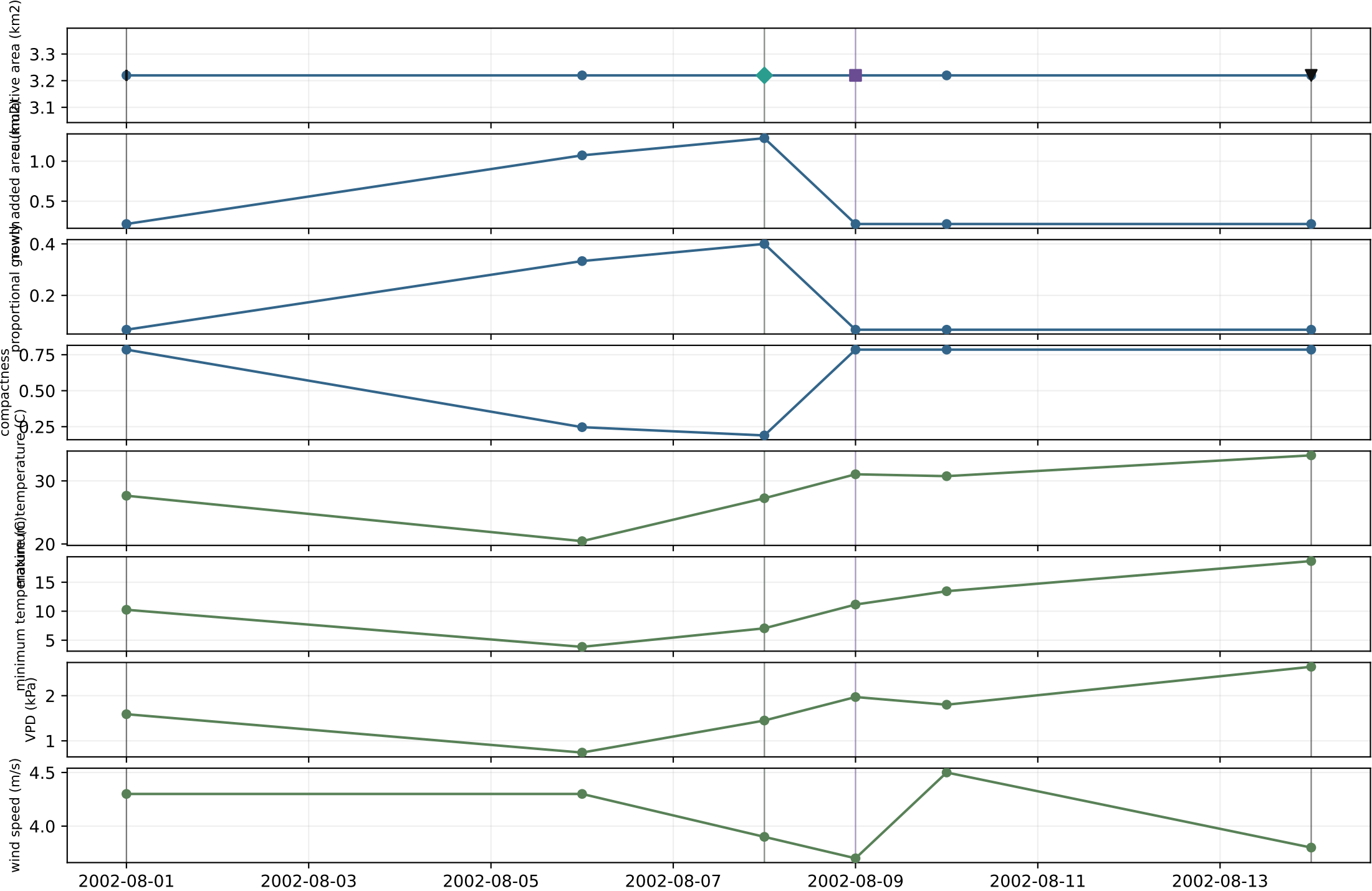
Fire 19752: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



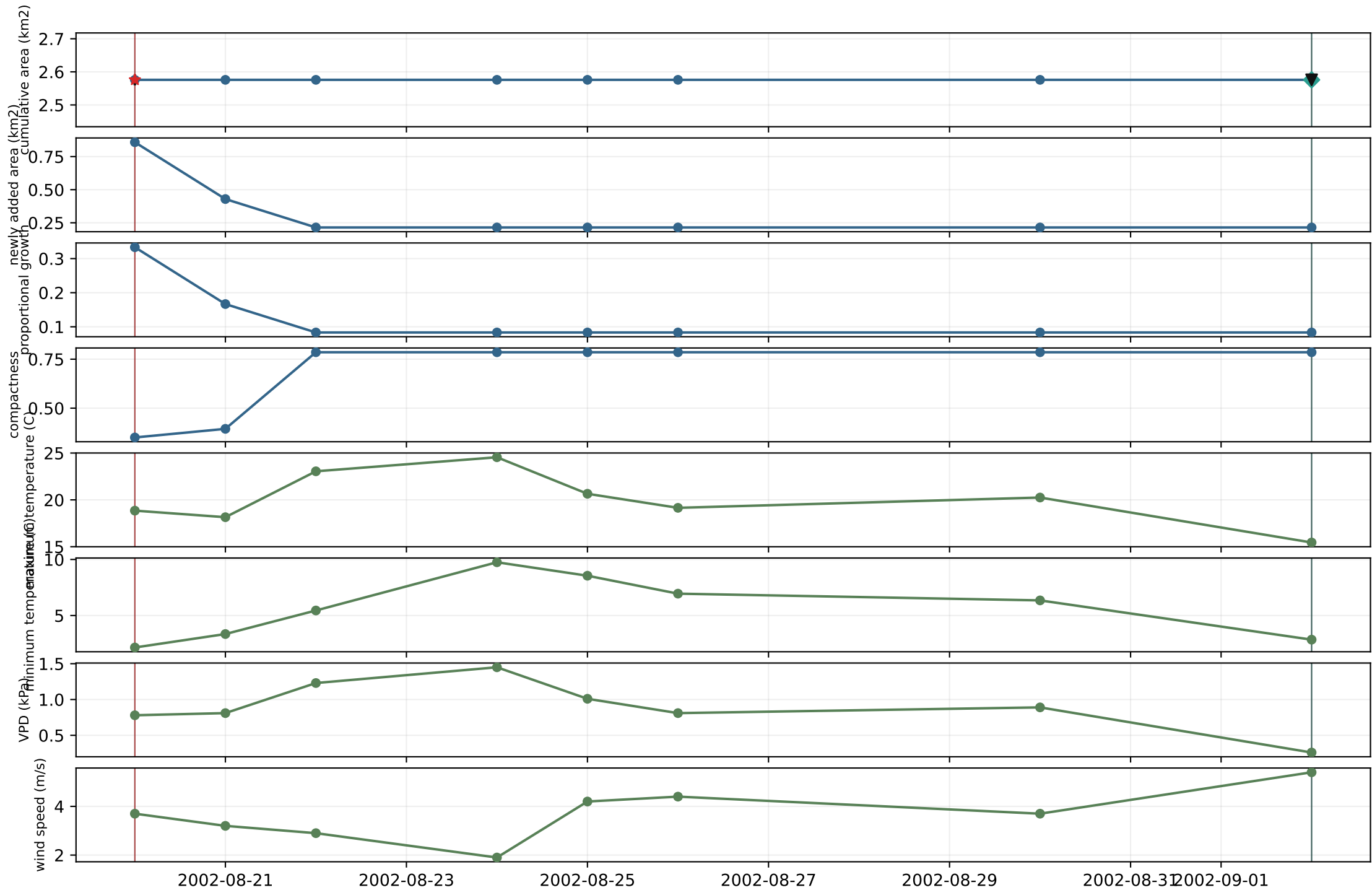
Fire 22518: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



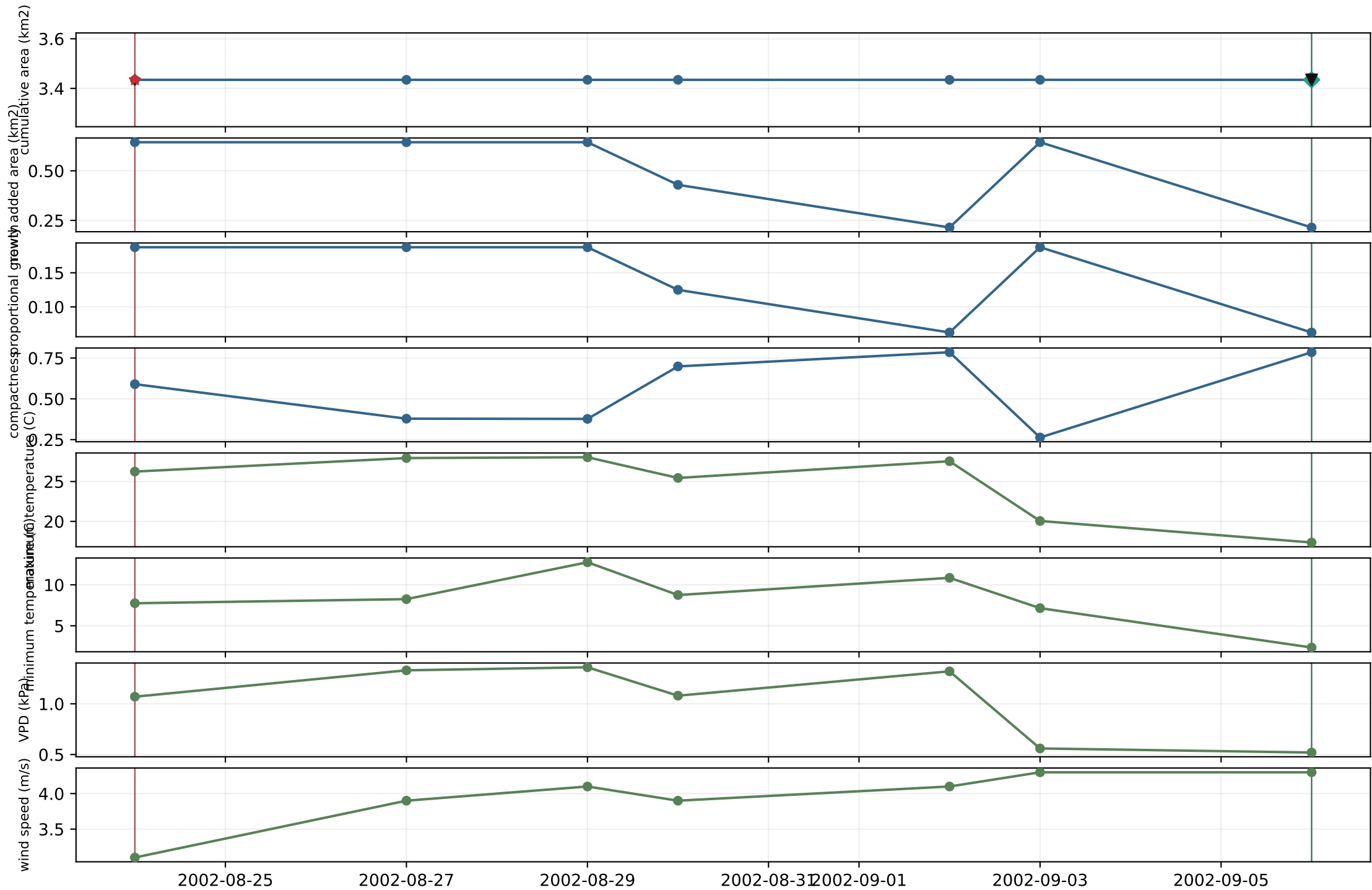
Fire 22615: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



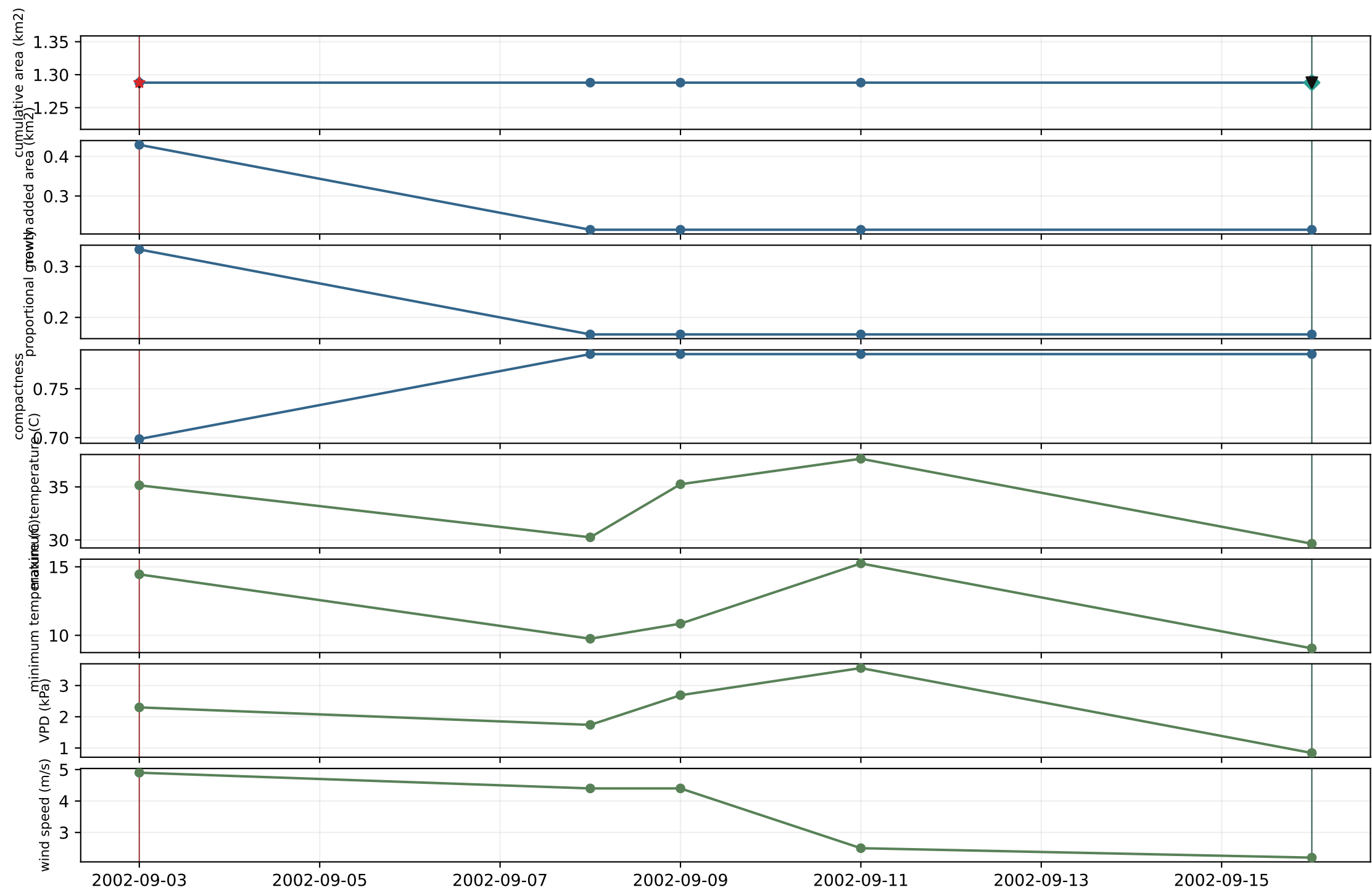
Fire 23942: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



Fire 24184: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction

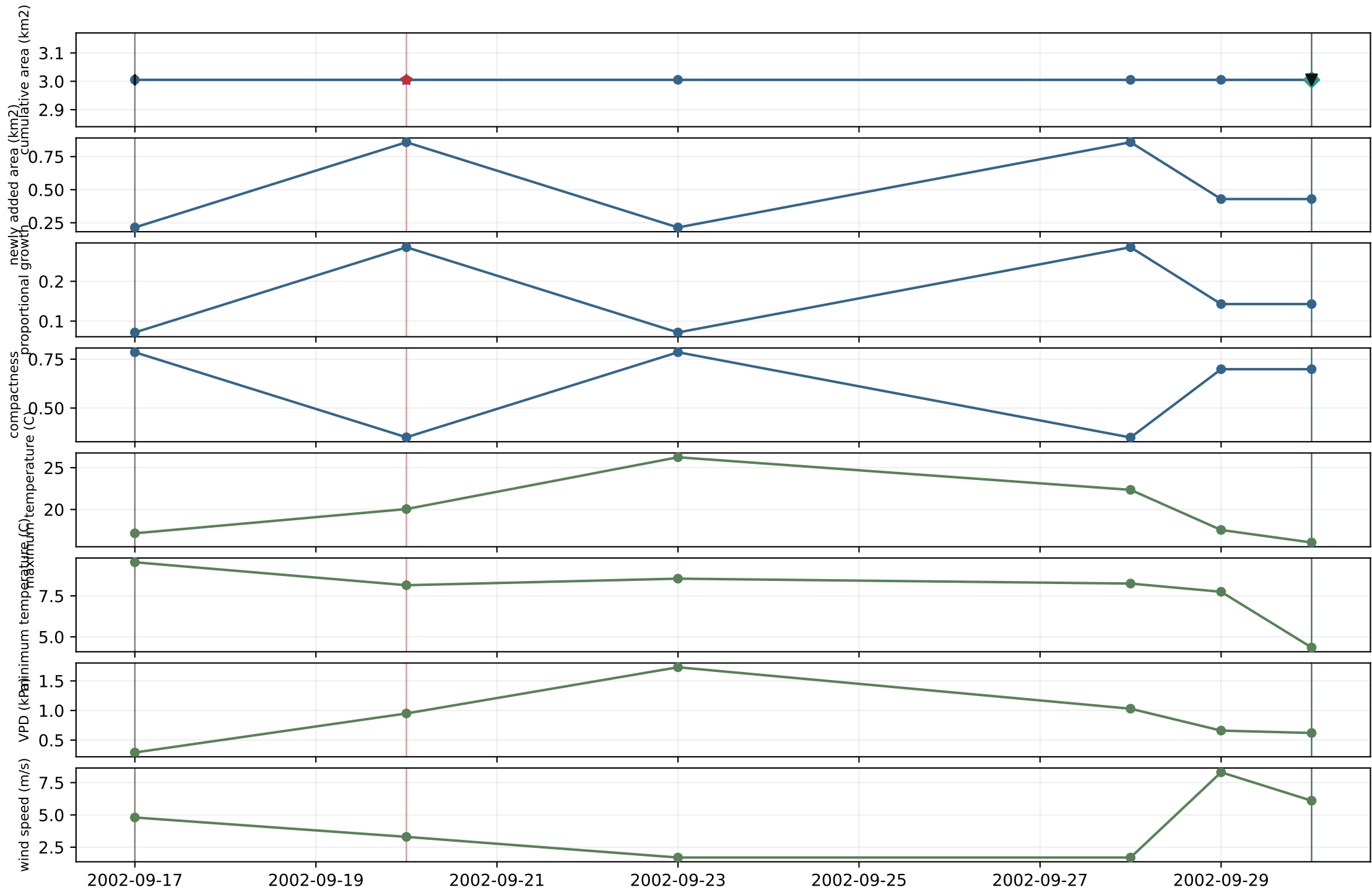


Fire 25068: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction

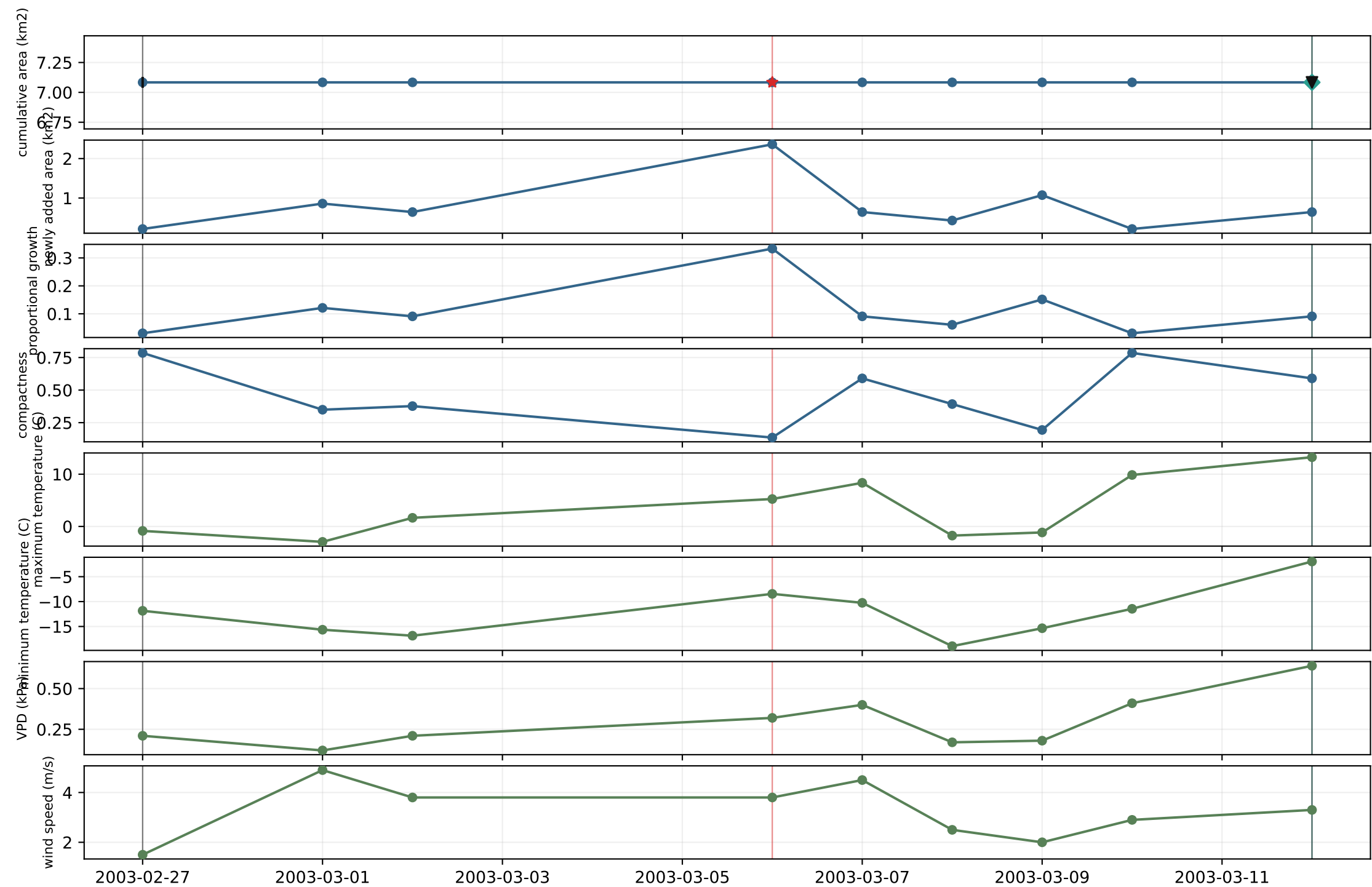


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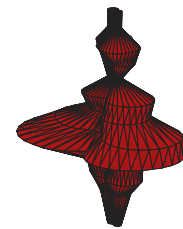
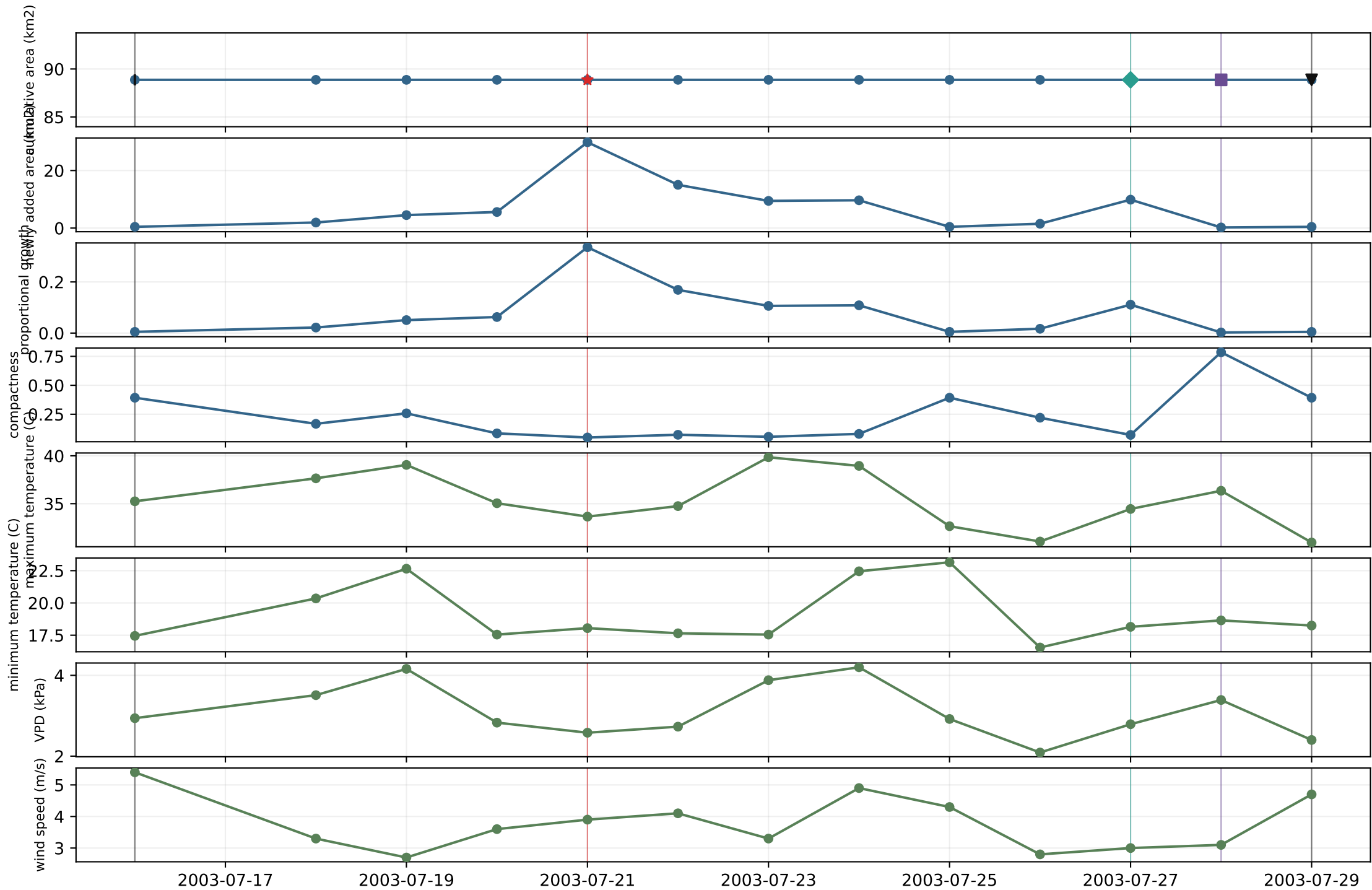
Fire 25790: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



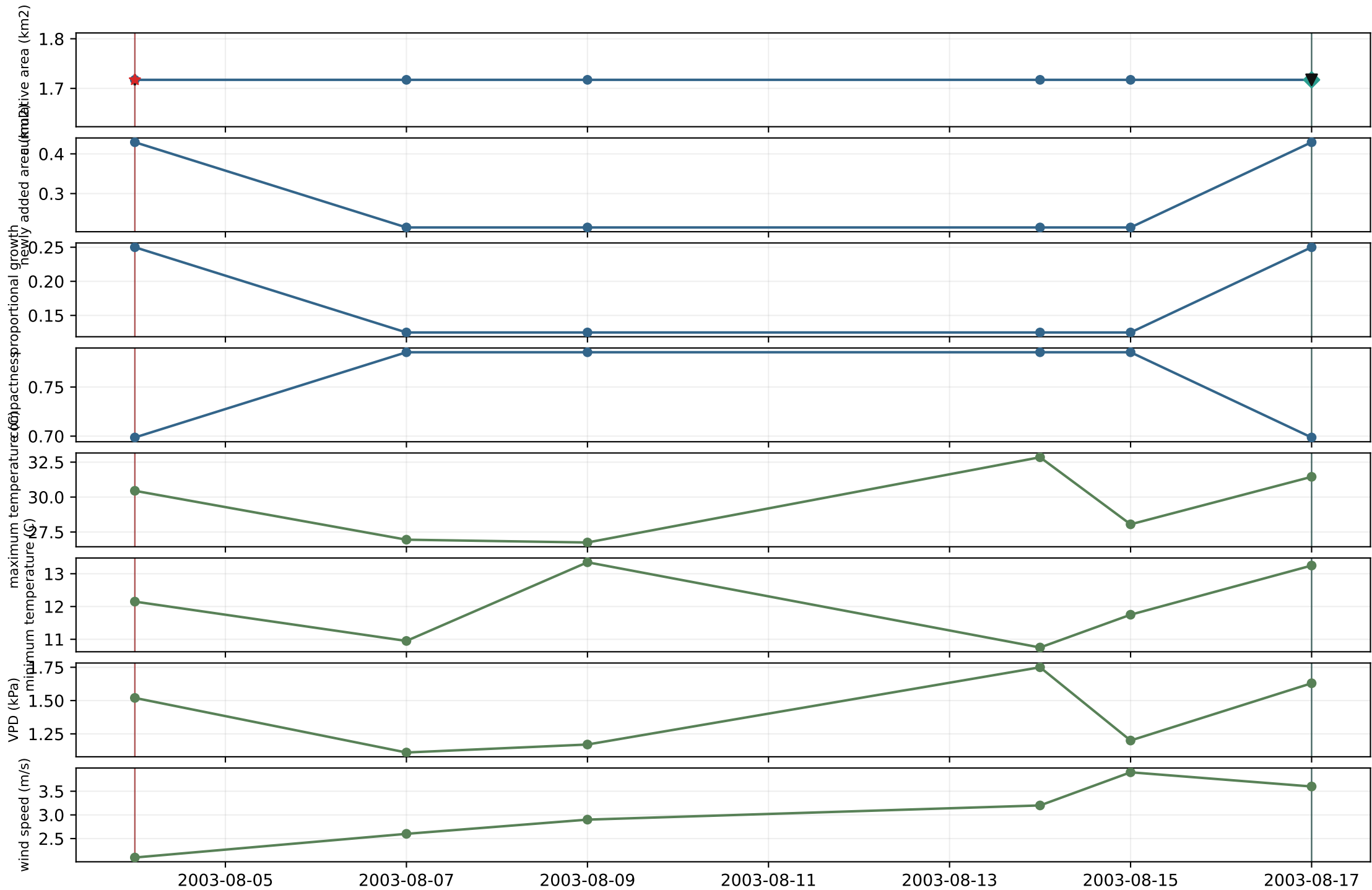
Fire 30922: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



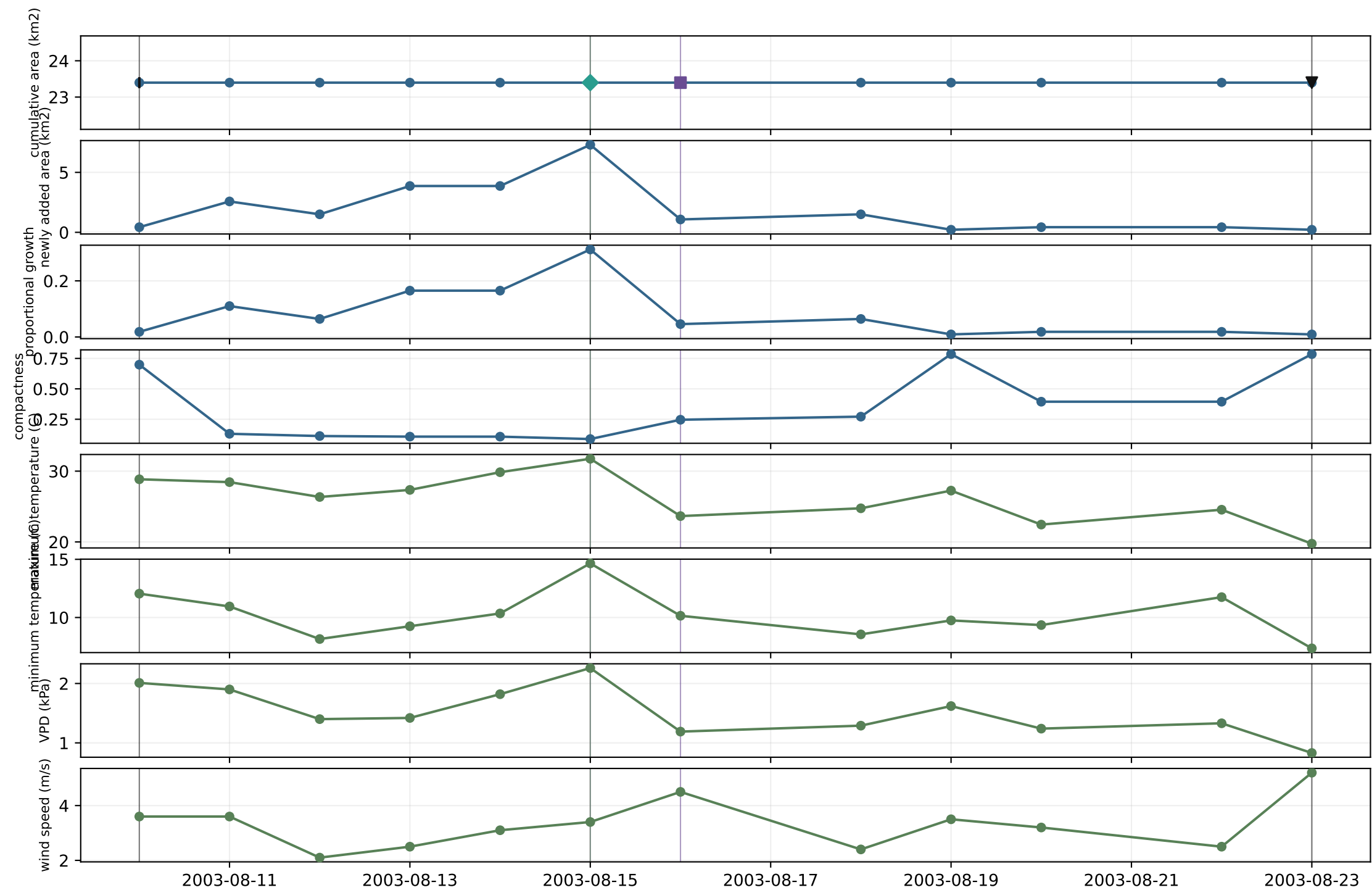
Fire 42013: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



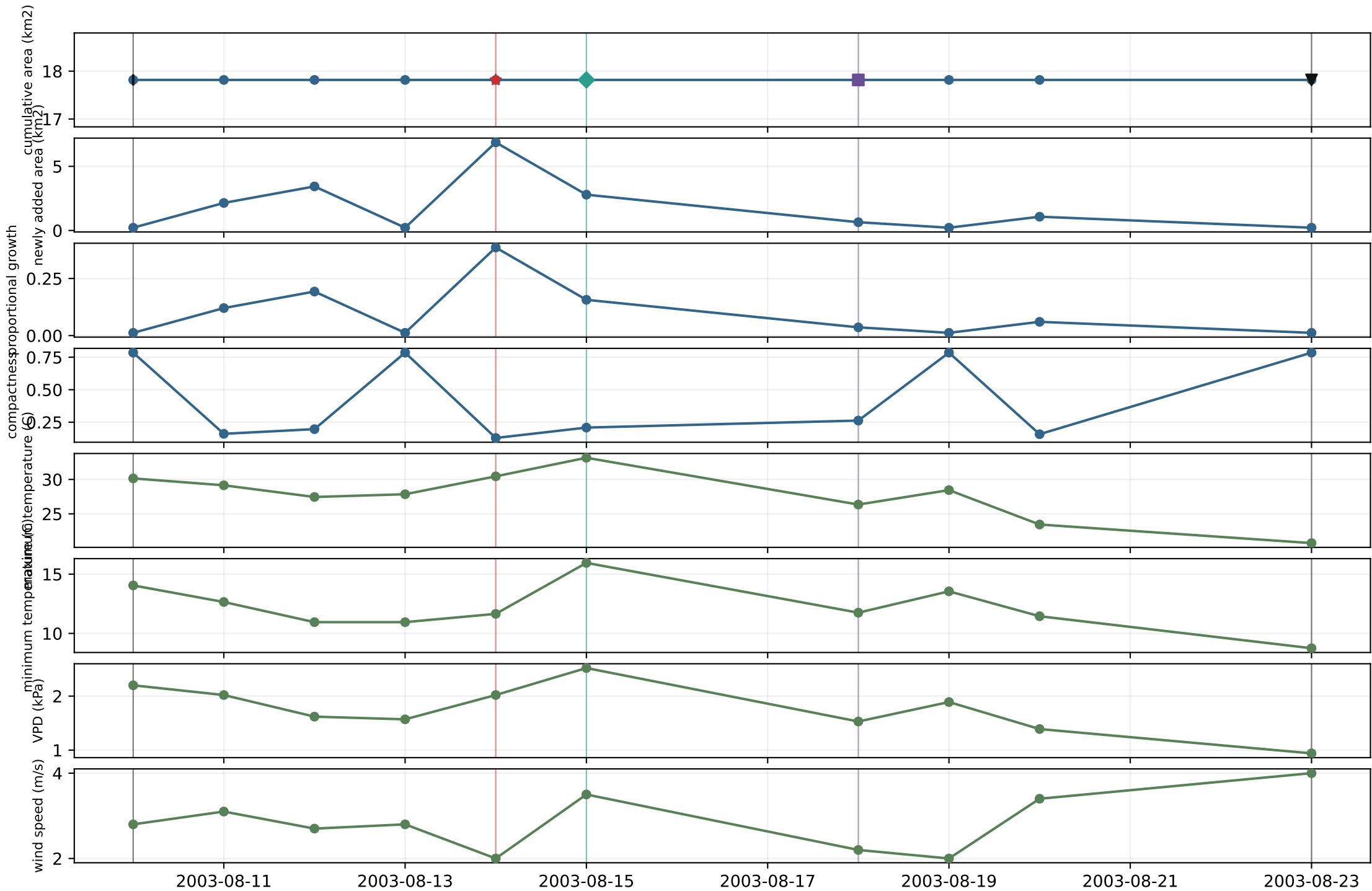
Fire 43422: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



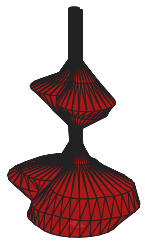
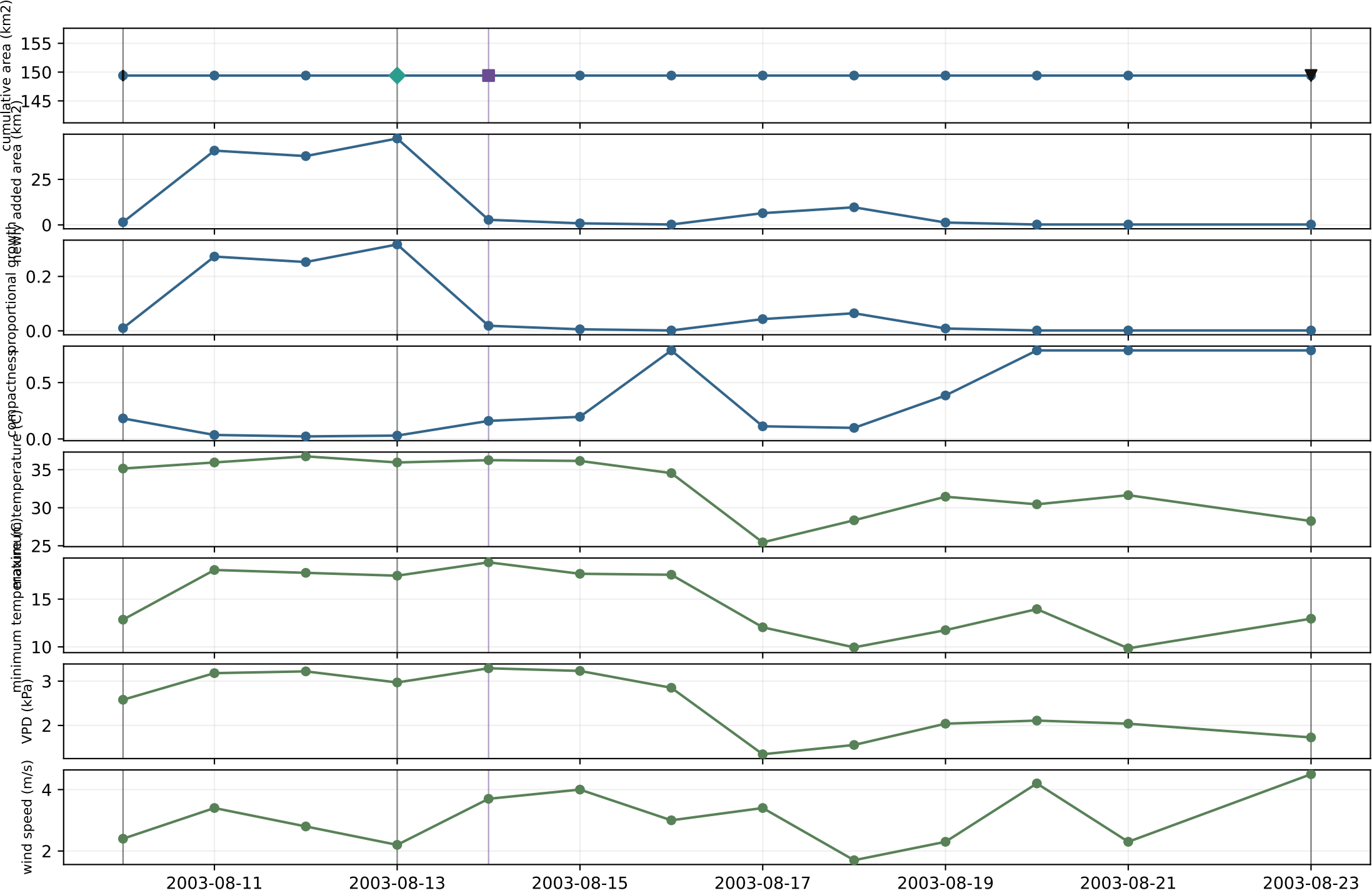
Fire 44170: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



Fire 44195: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



Fire 44221: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction



Fire 44389: observed developmental profile; final meaningful expansion != terminal observed record != physical extinction

